The Helix–Hilbert–Polya Bridge: A Canonical Source Proof

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Abstract

This paper gives the source-side helix proof in canonical mathematical form. The construction starts from the prime-generated $\pi/3$ helix, the associated unitary phasor flow, and the Gram–von Neumann self-adjoint operator. Zero capture is read through the resolvent trace $-L'/L$, with multiplicity given by the logarithmic derivative residue. A crossing is a confluence event: the fiber phase cancels, the amplitude reaches its threshold, the sign flips, and the next spectral harmonic is created. The Intermediate Value Theorem is used only to pass from a sign change of the continuous standing readout to an actual crossing. The midpoint comes from the source midline and no-drift projection: in the source chart that midpoint is $\pi/6$ in the $\pi/3$ chart; projection to the unit-circle readout preserves the midpoint, and the logarithmic one-dimensional readout is $1/2$. Thus the real coordinate $1/2$ is a geometric half-unit forced by the source crossing. The exponential source coordinate, the two-dimensional angle, and the final channel ordinate are distinct records of the same event; the final imaginary coordinate is the channel ordinate produced by the readout, not a number inserted as the three-dimensional source coordinate. The construction is unconditional: the prime-generated helix, the unitary phasor flow, the Gram–von Neumann self-adjoint operator, the no-radial-drift normal form, the Euler-product discriminator, and the resolvent-trace multiplicity are all established directly. On this foundation the zero-side pole, the source crossing, and the spectral harmonic are identified as a single event. The main result is the Hilbert–Polya closure: every Dirichlet-channel non-trivial zero is represented by the source spectral correspondence, every such spectral coordinate has real part $1/2$, and the principal zeta/L1 channel follows by eta regularization.

Contents

1 Objects and Notation

Let χ be a Dirichlet character modulo q and write

$$L(s,\chi)$$

for the associated Dirichlet L-function. For a channel χ, define the strip zero set

$$\mathrm{NTZ}(\chi) := \{\rho \in \mathbb{C} : 0 < \operatorname{Re}\rho < 1,\ L(\rho,\chi) = 0\}.$$

For a channel χ, write

$$\mathrm{GRH}(\chi) := \forall \rho \in \mathrm{NTZ}(\chi), \quad \operatorname{Re}\rho = \frac{1}{2}.$$

The all-channel closure statement is

$$\mathrm{GRH}_{\mathrm{HP}} := \forall q,\ \forall \chi \pmod q, \quad \mathrm{GRH}(\chi).$$

The principal zeta/L1 channel is treated separately through eta regularization. For zeta, let

$$\mathrm{NTZ}_\zeta := \{\rho \in \mathbb{C} : 0 < \operatorname{Re}\rho < 1,\ \zeta(\rho) = 0\}.$$

The zeta/L1 closure statement is

$$\mathrm{RH} := \forall \rho \in \mathrm{NTZ}_\zeta, \quad \operatorname{Re}\rho = \frac{1}{2}.$$

2 State of the Art and Contribution

The classical analytic theory begins with Riemann's use of $\zeta(s)$, its functional equation, and its zero set in the explicit study of primes [Rie59, Edw74, Tit86]. For Dirichlet and related L-functions, the standard arithmetic input is the Euler product, its logarithmic derivative, and the von Mangoldt prime-power trace [Dav00, MV07, IK04]. In this paper those facts are used as the source discriminator: the fiber being projected is the prime-generated Euler-product fiber rather than a function selected only by a matching functional equation.

The spectral viewpoint is usually formulated as a search for an operator or trace formula whose spectral side reads the critical zeros. Representative models include the $H = xp$ program, noncommutative trace formulas, and random matrix models for zero statistics and central values [BK99, Con99, KS00]. The present construction uses a different source object. The operator side is the Gram–von Neumann normal form $G_\infty = B_\infty^* B_\infty$ on the source Hilbert space; the geometric side is the prime-generated $\pi/3$ helix with no radial drift.

The Davenport–Heilbronn phenomenon shows why a functional-equation-level comparison is insufficient for this proof strategy [DH36]. The readout chain therefore includes an Euler-product discriminator before the Pythagorean midpoint step. An object lacking the prime-generated source fails that discriminator step and leaves the projection chain.

The contribution of this paper is the following canonical bridge:

(C1) construct the placed source fiber directly from the $\pi/3$ helix and the prime-generated channel data;

(C2) identify the closed-form amplitude accumulation $C^{-s}L(s,\chi)$ and the resolvent trace $C^{-s}(-L'/L)(s,\chi)$ as two readouts of the same source;

(C3) model zero capture as a crossing confluence where phase cancellation, amplitude threshold, sign flip, trace residue, and new harmonic creation occur with the Intermediate Value Theorem supplying the crossing from the sign change simultaneously;

(C4) prove that the source midline, no-drift condition, and faithful projection force the midpoint $\pi/6$ in the $\pi/3$ chart, and that the logarithmic one-dimensional readout is $(3/\pi)(\pi/6) = 1/2$.

(C5) identify every channel zero with the corresponding source spectral event and conclude $\mathrm{GRH}_{\mathrm{HP}}$, with the zeta/L1 channel obtained by the same eta-regularized readout.

3 Proof Roadmap

The proof has five layers.

(P1) **Source construction.** The integer fibers are placed on the $\pi/3$ helix by the area law $R_n^2 = Cn$, with $C = (\pi/3)^2$. Unique factorization supplies the prime coordinates, the channel weights, and the Euler-product discriminator.

(P2) **Trace and accumulation.** The placed fiber is the amplitude accumulation $C^{-s}L(s,\chi)$, while the resolvent trace is the prime-power accumulation $C^{-s}(-L'/L)(s,\chi)$. A captured zero is therefore read as a pole of the resolvent trace, with residue equal to its multiplicity up to the nonzero geometric gauge.

(P3) **Crossing geometry.** At a source crossing the channel phase cancels, the amplitude reaches a threshold, the sign flips, and the next harmonic begins. The Intermediate Value Theorem converts that sign change into an actual crossing; the source midline and no-drift projection place the crossing on the midpoint axis, which is $\pi/6$ in the $\pi/3$ chart.

(P4) **Projection readout.** The three-dimensional crossing projects to the unit circle without changing the midpoint angle. The logarithmic one-dimensional readout sends $\pi/6$ to $(3/\pi)(\pi/6) = 1/2$ while the lifted phase retains the height.

(P5) **HP closure.** The resolvent pole, source crossing, spectral harmonic, and projected one-dimensional zero are identified as one event. Self-adjointness makes the spectral ordinate real, and the spectral parametrisation $\rho = 1/2 + i\mu$ gives $\operatorname{Re}\rho = 1/2$.

The subsequent sections supply these layers in the order needed to reproduce the argument: source geometry, Euler-product generation, trace normal forms, standing readout, crossing costs, the R1–R12 infrastructure, the crossing definition, and finally the canonical closure.

4 The $\pi/3$ Helix Geometry

4.1 Canonical Scale

We use the canonical $\pi/3$ chart

$$A = \frac{\pi}{6}, \qquad ds = \frac{\pi}{3}, \qquad C = 2A\,ds = \left(\frac{\pi}{3}\right)^2.$$

The exact area-law placement is

$$R_n^2 = Cn, \qquad R_n = \frac{\pi}{3}\sqrt{n}.$$

Equivalently, the exponential source coordinate uses

$$x_n = \log R_n = \log\frac{\pi}{3} + \frac{1}{2}\log n, \qquad e^{x_n+iy} = R_n e^{iy}.$$

Thus the Euclidean radius is R_n, while the z-coordinate of the source chart is logarithmic in the radius. Measuring the source in $z = x+iy$ therefore uses e^{iy} for phase and e^z for the embedded helix point; it is not a linear vertical coordinate. At the source level the data are the $\pi/3$ coordinates, the channel weights coming from residue counting, and the harmonic phase on the helix. The fiber is derived from that source. No one-dimensional continuation formula is used to create it. The standard symbols L and ζ occur because the derived fiber identifies with those one-dimensional channel readouts after the geometric fiber has already been placed. Thus the three-dimensional source fiber atom at channel ordinate t, for $s = \frac{1}{2} + it$, is

$$a_n(s) = w(n)\frac{1}{R_n}\exp\{-it\operatorname{Log}(R_n^2)\} = w(n)(Cn)^{-s}.$$

In this notation the variable t is the logarithmic one-dimensional readout. The three-dimensional source height is exponential in that readout:

$$z_3(t) = e^t, \qquad z_1 = \log z_3 = t.$$

For a conical source helix

$$\Gamma(k) = \big(rk\cos(2\pi k),\, rk\sin(2\pi k),\, pk\big),$$

the event at readout t has

$$k(t) = \frac{e^t}{p}, \qquad R(t) = rk(t) = \frac{r}{p}e^t.$$

Thus the 3D source is measured in $z_3 = e^t$, and the one-dimensional readout is obtained by taking the logarithm. The continuous geometric integer count is the arclength count

$$N(t; p, r, \Delta) = \frac{1}{\Delta} S\left(\frac{e^t}{p}; p, r\right), \qquad \Delta = \frac{\pi}{3},$$

with

$$S(k; p, r) = \int_0^k \sqrt{p^2 + r^2 + (2\pi r u)^2}\, du.$$

The two-dimensional record is an angle. Keeping these three coordinates separate is part of the construction.

4.2 Closed-Form Geometry and Phasor Fiber

The conical helix used for the source geometry is

$$\gamma(k) = \big(rk\cos(2\pi k),\, rk\sin(2\pi k),\, pk\big).$$

At channel readout ordinate y the three-dimensional source height is

$$z_3 = e^y, \qquad k(y) = \frac{e^y}{p}, \qquad R(y) = rk(y) = \frac{r}{p}e^y,$$

so the same source event has the placed coordinate

$$\gamma(y) = \left(\frac{r}{p}e^y \cos\frac{2\pi e^y}{p}, \frac{r}{p}e^y \sin\frac{2\pi e^y}{p}, e^y\right).$$

The integer embedding is measured in fixed source cost

$$s_n = n\Delta, \qquad \Delta = \frac{\pi}{3}.$$

The closed arclength formula is

$$S(k; p, r) = \frac{k}{2}\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} + \frac{p^2 + r^2}{4\pi r}\sinh^{-1}\left(\frac{2\pi r k}{\sqrt{p^2 + r^2}}\right) \qquad (r > 0),$$

and therefore the source-consumption index is

$$N(y; p, r, \Delta) = \frac{1}{\Delta} S\left(\frac{e^y}{p}; p, r\right), \qquad N(y; p, r) = \frac{3}{\pi} S\left(\frac{e^y}{p}; p, r\right) \quad (\Delta = \pi/3).$$

Thus the cost between crossings is a fixed $\pi/3$ source increment, with the first crossing occurring at the half-cost $\pi/6$; the amount of channel height needed to earn that cost is determined by the fiber and its conductor.

For a Dirichlet character χ modulo q, the critical-line phasor fiber is

$$C_\chi(y) = \sum_n \chi(n)A(n)e^{-iy\log n}, \qquad A(n) = n^{-1/2}.$$

Formally,

$$C_\chi(y) = \sum_n \chi(n)n^{-1/2}e^{-iy\log n},$$

and the source cancellation condition is

$$C_\chi(y) = 0.$$

For complex characters,

$$\chi(n) = |\chi(n)|e^{i\arg\chi(n)}, \qquad v_n(y) = |\chi(n)|n^{-1/2}e^{i(\arg\chi(n)-y\log n)}.$$

For the principal character modulo q,

$$\chi_0(n) = \begin{cases} 1, & (n,q)=1, \\ 0, & (n,q)>1, \end{cases} \qquad L(s,\chi_0) = \zeta(s)\prod_{\ell|q}(1-\ell^{-s}).$$

The finite phasor chain records how the channels cancel at a placed source event. It is a diagnostic of the source balance. The continued evaluator is the corresponding L-function or eta-regularized principal readout below.

Lemma 4.1 (Area-law closed form)**.** *For any channel weights $w(n)$ for which the corresponding Dirichlet series is defined,*

$$F_w(s) := \sum_{n\geq 1} w(n)(Cn)^{-s} = C^{-s}\sum_{n\geq 1} w(n)n^{-s}.$$

Proof. Use $R_n^2 = Cn$ and $s = \frac{1}{2} + it$:

$$R_n^{-1}e^{-it\,\mathrm{Log}(R_n^2)} = (R_n^2)^{-1/2}(R_n^2)^{-it} = (R_n^2)^{-s} = (Cn)^{-s}.$$

Summing gives the result. □

4.3 Non-principal Character Channels

For a non-principal Dirichlet character χ modulo q, the channel weights are

$$w(n) = \chi(n).$$

Therefore the source channel sum is

$$F_\chi(s) = \sum_{n\geq 1}\chi(n)(Cn)^{-s}.$$

In the projected one-dimensional notation this same harmonic sum is

$$F_\chi(s) = C^{-s}L(s,\chi).$$

The channel has the finite conductor closed form

$$L(s,\chi) = q^{-s} \sum_{a=1}^{q} \chi(a)\, \zeta\left(s, \frac{a}{q}\right).$$

On the critical line $s = \frac{1}{2} + iy$ this reads

$$L\left(\frac{1}{2} + iy, \chi\right) = q^{-1/2} e^{-iy \log q} \sum_{a=1}^{q} \chi(a)\, \zeta\left(\frac{1}{2} + iy, \frac{a}{q}\right).$$

If $2 \nmid q$, the eta-like acceleration is

$$L_{\chi}^{(\eta)}(s) = \sum_{n \geq 1} (-1)^{n-1} \chi(n) n^{-s},$$

with

$$L_{\chi}^{(\eta)}(s) = (1 - 2^{1-s}\chi(2)) L(s,\chi), \qquad L(s,\chi) = \frac{L_{\chi}^{(\eta)}(s)}{1 - 2^{1-s}\chi(2)}.$$

For χ_3,

$$\chi_3(1) = 1, \qquad \chi_3(2) = -1, \qquad \chi_3(0) = 0,$$

and hence

$$F_{\chi_3}(s) = C^{-s} L(s, \chi_3).$$

4.4 The Principal/Zeta Channel

For the zeta/L1 channel the fiber is eta-regularized:

$$w(n) = (-1)^{n+1}.$$

The source channel sum is

$$F_{L1}(s) = \sum_{n \geq 1} (-1)^{n+1} (Cn)^{-s}.$$

In the projected one-dimensional notation the eta factor is

$$\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s} = (1 - 2^{1-s}) \zeta(s).$$

Thus the L1 fiber is

$$F_{L1}(s) = C^{-s} \eta(s) = C^{-s} (1 - 2^{1-s}) \zeta(s).$$

For the principal character modulo q,

$$L(s, \chi_0) = \frac{\eta(s)}{1 - 2^{1-s}} \prod_{\ell \mid q} (1 - \ell^{-s}).$$

At $s = \frac{1}{2} + iy$,

$$\eta\left(\frac{1}{2} + iy\right) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-1/2} e^{-iy \log n}.$$

On the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta factor is nonzero:

$$1 - 2^{1-s} \neq 0,$$

because $2^{1-s} = 1$ would imply $2^{1-\operatorname{Re} s} = 1$, hence $\operatorname{Re} s = 1$.

4.5 The Midpoint as the Reflection-Self-Dual Radius of the Channel

The midpoint $\sigma = 1/2$ is not inserted as an assumption; it is the unique radius at which the accumulated character channel is invariant under the reflection $s \mapsto 1-s$. The fiber is the sequence of partial sums of the signed Dirichlet series: one term

$$a_n(s) = \chi(n)\,(Cn)^{-s}$$

is incorporated for each integer n, with sign given by the character value. For a real (quadratic) channel the index set splits into a positive part, a negative part, and the part annihilated by χ,

$$\chi(n) = +1, \qquad \chi(n) = -1, \qquad \chi(n) = 0,$$

and the fiber is the difference of the two accumulated subseries,

$$F_\chi(s) = \sum_{n\geq 1} \chi(n)(Cn)^{-s} = P_+(s) - P_-(s), \qquad P_\pm(s) = \sum_{\chi(n)=\pm 1} (Cn)^{-s}.$$

The balance $P_+(\rho) = P_-(\rho)$ is precisely the vanishing $F_\chi(\rho) = 0$ of the channel.

Each index contributes a term at its own scale. The area law $R_n^2 = Cn$ makes the squared radius linear in n, so with

$$c_n = Cn = R_n^2 > 1$$

the n-th term is $a_n(s) = \chi(n)\,c_n^{-s}$, and by the no-radial-drift normal form below its modulus depends only on the abscissa,

$$|a_n(s)| = c_n^{-\sigma}, \qquad s = \sigma + iy,$$

independently of the ordinate y.

The completed source is two-sided (the double-ended helix below): the principal orientation and the reflected anti-helix realize the involution $s \mapsto 1-s$, pairing the n-th term with $c_n^{-(1-s)}$. The resulting reflection pair

$$Q_n(s) = c_n^{-s} + c_n^{-(1-s)}$$

has moduli $c_n^{-\sigma}$ and $c_n^{-(1-\sigma)}$, so any cancellation $Q_n(s) = 0$ forces

$$c_n^{-\sigma} = c_n^{-(1-\sigma)};$$

since $c_n > 1$, the map $u \mapsto c_n^u$ is strictly monotone, whence

$$-\sigma = -(1-\sigma) \qquad \Longrightarrow \qquad \sigma = \tfrac{1}{2}.$$

This conclusion is independent of c_n, so a single abscissa $\sigma = 1/2$ balances every reflection pair Q_n simultaneously: it is the unique reflection-self-dual radius of the whole channel.

The balance is attained there. At $s = \frac{1}{2} + i\pi/(2\log c_n)$ the two terms have equal modulus $c_n^{-1/2}$ and opposite phase,

$$c_n^{-s} = c_n^{-1/2} e^{-i\pi/2}, \qquad c_n^{-(1-s)} = c_n^{-1/2} e^{i\pi/2},$$

so $Q_n(s) = 0$.

Hence $\sigma = 1/2$ is the unique abscissa at which the channel is reflection-balanced, forced by the area-law scales, the absence of radial drift, and the two-sided reflection symmetry—not by any single scale and not by a downstream hypothesis on the zero set. Whether the actual balance $P_+(\rho) = P_-(\rho)$ occurs at this self-dual abscissa rather than at a functional-equation impostor is settled separately by the Euler-product discriminator below, which is what distinguishes the prime-generated source from a Davenport–Heilbronn object sharing only the reflection symmetry.

4.6 FTA and Euler Product Generation

The source helix is prime-generated. Its integer fibers are generated from prime coordinates by unique factorization, and the Euler product is the convergence-half-plane readout of that multiplicative geometry [Dav00, MV07].

Proposition 4.2 (FTA helix character)**.** *Fix a prime-angle assignment $\theta : \mathbb{N} \to \mathbb{R}$ and define the FTA winding angle*

$$\Theta_\theta(n) = \sum_{p^e \| n} e\,\theta(p).$$

For $m, n \geq 1$,

$$\Theta_\theta(mn) = \Theta_\theta(m) + \Theta_\theta(n).$$

Consequently the unit-circle winding

$$\omega_\theta(n) = \exp(i\Theta_\theta(n)),$$

is multiplicative:

$$\omega_\theta(mn) = \omega_\theta(m)\omega_\theta(n).$$

For a Dirichlet channel χ, the channel helix character

$$\Psi_{\chi,\theta}(n) = \chi(n)\omega_\theta(n)$$

is multiplicative:

$$\Psi_{\chi,\theta}(mn) = \Psi_{\chi,\theta}(m)\Psi_{\chi,\theta}(n) \qquad (m, n \geq 1).$$

Proof. By unique factorization,

$$v_p(mn) = v_p(m) + v_p(n)$$

for every prime p. Summing $e\,\theta(p)$ over the finite factorization support gives the displayed additivity of Θ_θ. Exponentiating converts addition of angles into multiplication on the unit circle. Since Dirichlet characters are multiplicative, the product $\chi(n)\omega_\theta(n)$ is multiplicative as well. □

Proposition 4.3 (Log bridge and Euler product readout)**.** *The logarithm enters only at the external analytic bridge. With bridge angle*

$$\theta_\gamma(p) = \gamma \log p,$$

the FTA identity

$$\log n = \sum_{p^e \| n} e \log p$$

gives

$$\Theta_{\theta_\gamma}(n) = \gamma \log n, \qquad \omega_{\theta_\gamma}(n) = n^{i\gamma}.$$

In the convergence half-plane $\operatorname{Re} s > 1$*, the same multiplicativity is read as an Euler product:*

$$F_\chi^C(s) = \sum_{n \geq 1} \chi(n)(Cn)^{-s} = C^{-s} \prod_p \left(1 - \chi(p)p^{-s}\right)^{-1}.$$

Taking the logarithmic derivative gives the prime-power trace generated by the Euler product:

$$\mathcal{T}_\chi^C(s) = C^{-s} \sum_{n \geq 1} \Lambda(n)\chi(n)n^{-s} = C^{-s}\left(-\frac{L'}{L}(s, \chi)\right).$$

The winded prime-field bridge is

$$\sum_{n\geq 1} \chi(n)\Lambda(n)\omega_{\theta_\gamma}(n)n^{-s} = -\frac{L'}{L}(s - i\gamma, \chi), \qquad \operatorname{Re} s > 1.$$

Where the product representation converges, it also gives the Euler-product edge fact

$$\operatorname{Re} s \geq 1 \Longrightarrow L(s,\chi) \neq 0 \qquad (\chi \neq 1).$$

Proof. The identity

$$\log n = \sum_{p^e \| n} e \log p$$

is the logarithmic image of unique factorization. Substituting $\theta_\gamma(p) = \gamma \log p$ into the FTA angle gives $\Theta_{\theta_\gamma}(n) = \gamma \log n$, hence $\omega_{\theta_\gamma}(n) = n^{i\gamma}$. In $\operatorname{Re} s > 1$, absolute convergence allows the completely multiplicative integer series to factor into its prime Euler product. Applying the logarithmic derivative to the Euler product gives the von Mangoldt prime-power trace. Multiplication by $\omega_{\theta_\gamma}(n) = n^{i\gamma}$ shifts the trace from s to $s - i\gamma$.

□

Thus composites are generated by their prime coordinates through FTA. This is the structural distinction from a functional-equation-only construction. The product convergence is half-plane-limited; the FTA multiplicativity of the helix character itself is height-independent.

For the principal L1 channel, the same prime generation is present with $\chi = 1$. The working fiber is eta-regularized,

$$F_{L1}(s) = C^{-s}(1 - 2^{1-s})\zeta(s),$$

and the zero readout uses the gauged zeta resolvent trace

$$C^{-s}\left(-\frac{\zeta'}{\zeta}(s)\right).$$

The eta factor handles the principal-channel pole while prime generation remains the source.

4.7 Euler-Product Discriminator

The previous propositions supply a discriminator for the source object rather than a standalone zero-location argument. The logarithmic Euler product

$$\log \zeta(s) = \sum_p \sum_{k\geq 1} \frac{1}{k} p^{-ks}$$

is a convergence-half-plane identity. Its role in the closure is to certify that the source fiber being projected is the prime-generated Dirichlet/zeta fiber rather than an object selected only by a matching functional equation.

Definition 4.4 (Euler-product discriminator)**.** *For a non-principal Dirichlet channel χ, the Euler-product discriminator*

$$\mathcal{E}_\chi$$

is the following package:

(E1) the source fiber factors into the prime Euler product in $\operatorname{Re} s > 1$*:*

$$F_\chi^C(s) = C^{-s} \prod_p (1 - \chi(p)p^{-s})^{-1};$$

(E2) the winded prime-power trace is the shifted logarithmic derivative:

$$\sum_{n\geq 1} \chi(n)\Lambda(n)\omega_{\theta_\gamma}(n)n^{-s} = -\frac{L'}{L}(s - i\gamma, \chi), \qquad \operatorname{Re} s > 1;$$

(E3) the Euler-product edge gives nonvanishing on $\operatorname{Re} s \geq 1$*:*

$$\operatorname{Re} s \geq 1 \Longrightarrow L(s, \chi) \neq 0.$$

Proposition 4.5 (Euler product in the readout chain)**.** *The unit-circle readout invokes the Euler-product discriminator before it converts a projected source event into the midpoint conclusion. Concretely, the readout step takes as input the discriminator* $\mathcal{E}_\chi$*, the fiber-capture hypothesis* $\rho \in \mathrm{NTZ}(\chi)$*, the no-radial-drift normal form, and the unit-circle projection* $z_2(\rho)$*, and produces*

$$\|w(\rho)\|^2 = 1;$$

the unit-circle identity then yields

$$\operatorname{Re} \rho = \frac{1}{2}.$$

Proof. The discriminator $\mathcal{E}_\chi$ is assembled first, from the Dirichlet Euler product and the winded von Mangoldt trace; only afterwards does the projection furnish the unit-circle readout $z_2(\rho)$ and the identity $\|w(\rho)\|^2 = 1$. Consequently a model defined by a functional equation alone reaches this step only if it supplies the same package $\mathcal{E}_\chi$. A Davenport–Heilbronn object shares the reflection symmetry but lacks the Euler-product package, and therefore fails at $\mathcal{E}_\chi$; this is the falsifier the discriminator provides. □

5 Source Accumulation and Resolvent Trace

The placed fiber accumulation and the resolvent-trace accumulation are distinct but linked closed forms. This section records both normal forms before the standing readout and crossing geometry are introduced.

5.1 Polya Operator Normal Form

For a non-principal channel, the placed source fiber is

$$F_\chi^C(s) = \sum_{n\geq 1} \chi(n)(Cn)^{-s} = C^{-s}L(s, \chi).$$

This is the channel amplitude accumulation: it is the harmonic sum of the source fiber terms in the $\pi/3$ area chart.

Lemma 5.1 (No radial drift in the accumulation normal form)**.** *The helix source has no radial drift by construction. For each source atom*

$$a_n(t) = w(n)R_n^{-1} \exp\{-it \operatorname{Log}(R_n^2)\},$$

the radial size is independent of t:

$$|a_n(t)| = \frac{|w(n)|}{R_n} = \frac{|w(n)|}{(\pi/3)\sqrt{n}} \qquad (C = (\pi/3)^2).$$

Proof. For real t and positive R_n^2,

$$\left|\exp\{-it \operatorname{Log}(R_n^2)\}\right| = 1.$$

The helix flow therefore changes only phase. The radial coordinate is fixed by the source placement $R_n^2 = Cn$, so the accumulated source is no-drift in the radial direction before projection. □

This no-drift normal form is the invariant used by the Polya induction. Moving from one crossing to the next advances the phase ledger and harmonic threshold, not the radial placement. The same $\pi/3$ source geometry therefore persists through the infinite ladder of crossings. In the R1–R12 summary below, this is the source-side input for the geometric crossing correspondence and the resolvent-trace identification.

Definition 5.2 (Admissible source harmonic domain)**.** *For $s = \sigma + it$ and a finite set $F \subset \mathbf{N}$, the admissible finite source harmonic is*

$$H_{F,\chi}(s) = \sum_{n\in F} \chi(n)n^{-\sigma}U(t,n)^{-1}, \qquad U(t,n) = \exp(it \log n).$$

For the initial segment $F = \{0, \dots, M\}$ this is the finite source chain. The admissible source domain is the increasing ladder obtained from these finite source harmonics by threshold induction, with the $\pi/3$ coordinate chart applied at every crossing. No Green–Helmholtz energy norm is part of this source domain.

Proposition 5.3 (Source operator normal form)**.** *The Polya source flow is the diagonal unitary flow*

$$U(t,n) = \exp(it \log n), \qquad \|U(t,n)\| = 1,$$

with real generator

$$H(n) = \log n, \qquad \Im H(n) = 0.$$

Consequently a conserved nonzero source amplitude has zero radial drift.

Proof. The flow is unitary because for real t,

$$|e^{it \log n}| = 1$$

for every n. Its Stone generator is the real diagonal multiplication operator

$$H(n) = \log n.$$

Since the generator has no imaginary part and the flow is unitary, a conserved nonzero amplitude cannot acquire an exponential radial factor. The no-drift lemma then gives $\operatorname{Re}\lambda = 0$ for the source drift rate. □

Theorem 5.4 (The helix is its own continuation into the strip)**.** *For a non-principal channel χ, define the finite source phasor chain*

$$S_M(s,\chi) = \sum_{0\le n\le M} \chi(n) n^{-\sigma} U(t,n)^{-1}, \qquad s=\sigma+it, \quad U(t,n) = \exp(it\log n).$$

This is the same three-dimensional helix fiber, projected through the area-law/log bridge. For every s with $\sigma > 0$ there is a constant $C_{\chi,s} > 0$ such that

$$|S_M(s,\chi) - L(s,\chi)| \le C_{\chi,s} M^{-\sigma} \qquad (M \ge 1).$$

Thus the helix source chain continues itself from the convergent half-plane into the critical strip. At a zero ρ with $0 < \operatorname{Re}\rho < 1$,

$$|S_M(\rho,\chi)| \le C_{\chi,\rho} M^{-\operatorname{Re}\rho},$$

so the zero is read as a source fiber cancellation event. Moreover the rescaled chain exposes the height-free source ledger:

$$|S_M(\rho,\chi)M^\rho - (A_\chi(M) - L(0,\chi))| \le C'_{\chi,\rho} M^{-1}, \qquad A_\chi(M) = \sum_{0\le n\le M} \chi(n).$$

Proof. The phasor-flow form of the source chain is the displayed finite sum. The continuation estimate is obtained by summation by parts using the bounded partial sums of a non-principal character. At a zero, substituting $L(\rho,\chi) = 0$ gives the cancellation estimate. Multiplying the estimate by M^ρ gives the height-free ledger displayed above. This is not the divergent Euler product in the strip; it is the same harmonic source chain continued by the closure ledger. The zeta/L1 channel uses the parallel eta-regularized fiber and standing readout. ☐

5.2 Projection and Threshold Normal Forms

Proposition 5.5 (Pythagorean midpoint readout)**.** *For a projected source fiber event, the Pythagorean unit-circle readout forces*

$$\|w(\rho)\|^2 = 1 \iff \operatorname{Re}\rho = \frac{1}{2}.$$

At the $\pi/3$ source scale this is the same midpoint statement

$$\alpha(\operatorname{Re}\rho) = \frac{\pi}{6}.$$

Equivalently, if $z_2(\rho) \in S^1$ is the two-dimensional unit-circle projection of the crossing, written in the $\pi/3$ angular chart, then the one-dimensional readout is the lifted logarithm

$$\ell(\rho) = \frac{3}{\pi}\frac{1}{i}\operatorname{Log}_{\mathcal{H}} z_2(\rho), \qquad z_2(\rho) = \exp\Big(i\,\frac{\pi}{3}\ell(\rho)\Big),$$

where $\operatorname{Log}_{\mathcal{H}}$ is the continuous helix branch selected by the crossing ledger. Thus the unit-circle projection is still a $\pi/3$-coordinate object, and the 1D value is obtained by taking its logarithm and applying the same scale conversion.

Proof. The input here is the projected unit-circle datum, not the conclusion $\operatorname{Re}\rho = 1/2$. The source crossing supplies a no-drift unitary readout $z_2(\rho) \in S^1$, and the Möbius coordinate

$$w(\rho) = 1 - \frac{1}{\rho}$$

is the scalar chart assigned to that datum. Thus $\|w(\rho)\|^2 = 1$ is the Pythagorean statement that the projected readout lies on the unit circle. Since a nontrivial zero is not 0, the algebra is reversible:

$$|w(\rho)|^2 = 1 \iff \left(1 - \frac{1}{\rho}\right)\left(1 - \frac{1}{\overline{\rho}}\right) = 1 \iff \rho + \overline{\rho} = 1 \iff \operatorname{Re}\rho = \frac{1}{2}.$$

The $\pi/3$ coordinate conversion is

$$\alpha(x) = \frac{\pi}{3}x, \qquad \alpha^{-1}(u) = \frac{3}{\pi}u.$$

The value 1/2 is therefore the unit-readout midpoint forced by the source sign-change/no-drift crossing after source projection, not an independent condition on the zero set. The Pythagorean step uses only unit-circle geometry, the no-drift projection, and the Möbius chart; the midpoint is the derived coordinate value. □

Proposition 5.6 (Threshold ladder)**.** *The first source crossing has amplitude cost $\pi/2$, and each subsequent harmonic threshold advances by π:*

$$A_0^+ = \frac{\pi}{2}, \qquad A_{k+1}^+ = A_k^+ + \pi.$$

At a threshold $E(t) = n\pi$, the harmonic count is exactly n.

Proof. The positive ladder is $A_k^+ = (k + \frac{1}{2})\pi$, so $A_0^+ = \pi/2$ and $A_{k+1}^+ - A_k^+ = \pi$. If the harmonic count is defined by $\lfloor E(t)/\pi \rfloor$ at threshold, then $E(t) = n\pi$ gives count n. The channel cancellation law is the equality of the positive and negative parts of the signed channel sum. □

Remark 5.7 (Polya chain components)**.** *The operator used here is the Gram/von Neumann self-adjoint operator $G_\infty = B_\infty^* B_\infty$. Green–Helmholtz energy and finite Dirac operator models are separate analytic models. The Polya chain used here is exactly the Gram/von Neumann self-adjoint operator, phasor-flow unitarity, real generator/no-drift, threshold ladder, source cancellation, resolvent pole, $\pi/3$ chart, and Pythagorean readout path above.*

Proposition 5.8 (Exact multiplicity in the trace)**.** *Assume the boundary trace identity*

$$T(z) = -\frac{L'}{L}\left(\frac{1}{2} + iz, \chi\right).$$

If $\rho \in \mathrm{NTZ}(\chi)$ has order m, then at $z = z_\rho$ the resolvent trace has a simple principal part with residue $i\,m$. In the s-variable and $\pi/3$ gauge this is the same residue law

$$\operatorname*{Res}_{s=\rho} \mathcal{T}_\chi^C(s) = -mC^{-\rho}.$$

Proof. If $L(s,\chi) = (s-\rho)^m g(s)$ with $g(\rho) \neq 0$, then $L'/L = m/(s-\rho) + g'/g$. Under the boundary coordinate $s = \frac{1}{2} + iz$, the pole parameter changes the residue by the coordinate Jacobian, giving the displayed $i\,m$ in the z-trace. The multiplicity is therefore read from the Laurent residue of the resolvent trace as an exact local invariant. □

Proposition 5.9 (Source normal form for crossing correspondence). *The source normal form used by the crossing correspondence has three components. First, the radial drift rate vanishes under source conservation. Second, the threshold-height ladder satisfies*

$$E(h_n) = n\pi, \qquad \mathcal{N}_E(h_n) = n,$$

with rigidity for every sequence obeying the same threshold law. Third, each threshold crossing is identified with the corresponding midpoint zero and its unit-circle readout.

Proof. The no-radial-drift normal form fixes the placement $R_n^2 = Cn$ at every threshold. Continuity and strict monotonicity of the accumulation E locate each crossing, and rigidity of the threshold ladder shows that the threshold sequence (h_n) is the only crossing sequence. The $\pi/3$ chart supplies the midpoint coordinate, and the resolvent-pole side is supplied by the logarithmic-derivative pole law. □

5.3 Trace Residues and Completed Readout

The resolvent trace is the logarithmic, prime-power-weighted accumulation. In the convergent half-plane,

$$\begin{aligned}\mathcal{T}_\chi^C(s) &:= \sum_{n\ge1} \Lambda(n)\chi(n)(Cn)^{-s} \\ &= C^{-s}\sum_{n\ge1} \Lambda(n)\chi(n)n^{-s} \\ &= C^{-s}\left(-\frac{L'}{L}(s,\chi)\right).\end{aligned}$$

Here Λ is the von Mangoldt weight. Thus the same source geometry has two closed-form accumulations: the fiber sum $C^{-s}L(s,\chi)$ and the resolvent trace $C^{-s}(-L'/L)(s,\chi)$. At the canonical scale $C = (\pi/3)^2$, the continued trace is

$$\mathcal{T}_\chi^{\pi/3}(s) = \left(\left(\frac{\pi}{3}\right)^2\right)^{-s}\left(-\frac{L'}{L}(s,\chi)\right),$$

which is the canonical $\pi/3$ continued trace.

For numerical and closed-form readout, principal and non-principal channels are kept separate. If χ_0 is the principal character modulo q, then

$$L(s,\chi_0) = \zeta(s)\prod_{\ell\mid q}(1-\ell^{-s}) = \frac{\eta(s)}{1-2^{1-s}}\prod_{\ell\mid q}(1-\ell^{-s}),$$

where $\eta(s) = \sum_{n\ge1}(-1)^{n-1}n^{-s}$ denotes its analytic continuation. The alternating coprime-filter series

$$\sum_{(n,q)=1}(-1)^{n-1}n^{-s}$$

is therefore a separate accelerated readout: it equals $(1-2^{1-s})L(s,\chi_0)$ when q is odd, and equals $L(s,\chi_0)$ when $2\mid q$. If $2\nmid q$, the eta-like alternating acceleration satisfies

$$\sum_{n\ge1}(-1)^{n-1}\chi(n)n^{-s} = (1-2^{1-s}\chi(2))L(s,\chi),$$

while for $2 \mid q$ this factor degenerates to 1. These formulas are the continued evaluators; finite phasor cutoffs remain diagnostics for the placed source, not precision zero finders.

The literal logarithmic derivative of the placed fiber differs only by a regular gauge term:

$$-\frac{d}{ds}\log F_\chi^C(s) = \log C - \frac{L'}{L}(s,\chi).$$

The additional term $\log C$ has no pole. The multiplicative geometric gauge C^{-s} is never zero for $C > 0$. Therefore neither gauge changes the crossing locations or creates/removes trace poles; the multiplicative gauge only rescales the trace residue by a nonzero geometric factor.

This gives the local residue law at a capture event. If ρ is a zero of $L(s,\chi)$ of order m and

$$L(s,\chi) = (s-\rho)^m g(s), \qquad g(\rho) \neq 0,$$

then near ρ

$$-\frac{L'}{L}(s,\chi) = -\frac{m}{s-\rho} - \frac{g'}{g}(s),$$

and hence

$$\mathcal{T}_\chi^C(s) = -\frac{mC^{-\rho}}{s-\rho} + \text{a term finite at } \rho.$$

So the trace pole has residue $-mC^{-\rho}$. The integer m is the spectral multiplicity; the factor $C^{-\rho}$ is the nonzero $\pi/3$ geometric scaling. At a crossing, this is the residue consumed by the resolvent trace while the source-fiber amplitude attains its threshold, the channel phase cancels, and the sign changes. Since the standing readout is continuous, the Intermediate Value Theorem gives an actual crossing between the two signs. The Polya threshold condition is therefore the simultaneous event

$$F_\chi^C(\rho) = 0, \qquad \operatorname*{Res}_{s=\rho} \mathcal{T}_\chi^C(s) = -mC^{-\rho},$$
$$E_\chi(\operatorname{Im}\rho) = A_k^\pm, \qquad A_0^\pm = \pm\frac{\pi}{2}, \qquad A_{k+1}^\pm - A_k^\pm = \pm\pi.$$

Here E_χ is the source accumulation ledger used by the Polya chain. The jump from one level $A_k^\pm$ to the next is the threshold that starts the next harmonic on the spectral readout. Thus the closed-form resolvent residue, the crossing sign change, and the creation of the next harmonic are the same source event viewed in three ledgers: trace, standing wave, and harmonic ladder.

For primitive non-principal χ, the completed zero-side resolvent trace is the partial-fraction accumulation

$$\mathcal{R}_\chi(s) = \sum_{\rho \in \mathrm{NTZ}(\chi)} m_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho}\right),$$

where m_ρ is the vanishing order of the completed L-function at ρ. The closed form is

$$\frac{\Lambda_\chi'}{\Lambda_\chi}(s) = A + \mathcal{R}_\chi(s) \qquad (s \notin \mathrm{NTZ}(\chi)),$$

for a constant A. Splitting the completed function $\Lambda_\chi = \gamma_\chi L(s,\chi)$ gives, off the zeros in the strip,

$$-\frac{L'}{L}(s,\chi) = -A - \mathcal{R}_\chi(s) + \frac{\gamma_\chi'}{\gamma_\chi}(s).$$

Thus the prime-power accumulation, the zero-side resolvent trace, and the continued channel readout are the same readout up to the constant and Archimedean gauge terms, which are regular away from their own gauge poles.

For the L1/zeta channel,

$$F_{L1}^{C}(s) = C^{-s}(1-2^{1-s})\zeta(s).$$

Off $\operatorname{Re} s = 1$, the eta factor is nonzero, so a zero of the L1 fiber is a zero of ζ. The corresponding continued helix resolvent trace is

$$\mathcal{T}_{L1}^{C}(s) = C^{-s}\left(-\frac{\zeta'}{\zeta}(s)\right).$$

The eta factor regularizes the principal fiber; it is not inserted into the zero-pole readout, because the pole is the logarithmic derivative of the zeta factor whose zeros are being read. The raw trace is

$$-\frac{\zeta'}{\zeta}(s),$$

and the $\pi/3$ zeta fiber zero has both a raw and a gauged resolvent-trace pole, since multiplication by the nonzero gauge C^{-s} cannot remove a pole.

6 Standing Readout and Projection to One Dimension

This is the layer where the placed three-dimensional fiber is read in the lower-dimensional coordinate chart. The functions below are readouts of the source geometry and channel counts, with the coordinate phase correction kept explicit.

For $s = \frac{1}{2} + it$, the projected complex fiber is

$$F_\chi(t) = F_\chi\left(\frac{1}{2} + it\right).$$

The real standing readout has the form

$$Z_\chi(t) = \operatorname{Re}\left(e^{i\Theta_\chi(t)} F_\chi\left(\frac{1}{2} + it\right)\right),$$

where

$$\Theta_\chi(t) = \arg\Gamma\left(\frac{1/2 + a}{2} + \frac{it}{2}\right) + \frac{t}{2}\log\frac{q}{\pi} + t\log C + \alpha_\chi.$$

Here $a \in \{0,1\}$ is the parity and α_χ is the character/root-number gauge. For the L1/zeta eta-regularized channel one subtracts the eta gauge

$$\arg(1-2^{1-s}).$$

The term $t\log C$ is not a new oscillation. It is the coordinate correction from

$$\operatorname{Log}(R_n^2) = \operatorname{Log}(Cn) = \log n + \log C.$$

Omitting it rotates the real readout into the wrong coordinate chart.

6.1 Phase Conventions

All logarithms of positive real quantities are real logarithms. Thus

$$\mathrm{Log}(R_n^2) = \log(Cn), \qquad \log q, \qquad \log C$$

use the real branch. The gamma phase in the standing readout is the continuous phase

$$\vartheta_\chi(t) = \operatorname{Im}\log\Gamma\left(\frac{1/2 + a + it}{2}\right) + \frac{t}{2}\log\frac{q}{\pi},$$

with the branch chosen continuously along the vertical line $z(t) = (1/2+a+it)/2$. The root-number phase is fixed by

$$\tau(\chi) = \sum_{r=1}^{q} \chi(r) e^{2\pi i r/q}, \qquad \varepsilon_\chi = \frac{\tau(\chi)}{i^a\sqrt{q}}, \qquad \alpha_\chi = -\frac{1}{2}\arg(\varepsilon_\chi).$$

For the principal zeta/L1 channel, $\alpha_\chi = 0$ and the two-channel eta readout uses the continuous phase

$$\phi_\eta(t) = \arg\left(1 - 2^{1-(1/2+it)}\right),$$

so that

$$\Theta_{L1}(t) = \vartheta_\zeta(t) + t\log C - \phi_\eta(t).$$

For non-principal χ,

$$\Theta_\chi(t) = \vartheta_\chi(t) + t\log C + \alpha_\chi.$$

7 Harmonic Cost and the Threshold Ladder

The full source geometry is double ended. The midpoint is the origin of the fiber, and the reflected side is an anti-helix carrying the opposite orientation. The proof is written on the positive oriented side of the helix; the reflected anti-helix supplies the symmetric pre-origin geometry used for high-precision tail construction before the counting ledger starts at the origin. Its cost law is the same with the sign reversed.

The crossing levels in the projected unit-circle readout are therefore

$$A_k = \left(k + \frac{1}{2}\right)\pi, \qquad k \in \mathbb{Z}.$$

Equivalently, for $m \geq 0$ the two oriented ladders are

$$A_m^+ = \left(m + \frac{1}{2}\right)\pi, \qquad A_m^- = -\left(m + \frac{1}{2}\right)\pi.$$

On the positive helix side used in the proof,

$$A_0 = \frac{\pi}{2}, \qquad A_{k+1} = A_k + \pi.$$

Thus the first crossing is the only crossing on that side whose cost from the origin is the half-step $\pi/2$:

$$|A_0^+| = \frac{\pi}{2}.$$

Every additional crossing on that same side costs one full step π:

$$A^+_{m+1} - A^+_m = \pi \qquad (m \geq 0).$$

The reflected anti-helix obeys the same rule:

$$|A^-_0| = \frac{\pi}{2}, \qquad |A^-_{m+1} - A^-_m| = \pi \qquad (m \geq 0).$$

So the half-step $\pi/2$ occurs once, at the first crossing on each oriented side of the double-ended helix; all subsequent crossings advance by the full step π.

8 Hilbert–Polya Source Infrastructure: R1–R12

This section lists the mathematical infrastructure used by the proof. R1–R11 are the operator, flow, source, zero, crossing, coordinate, and spectral-reality facts. R12 is the counting and midpoint-identification package used to close the projection readout. The geometric projection proves that the three-dimensional midline projects to the one-dimensional midpoint 1/2. The origination bridge identifies the source mechanism as the prime-built helix and the one-dimensional zeros as dependent zeros of the zeta/L-function trace. The faithfulness bridge records the reconstruction bijection and injective log-readout: the retained channel plus the loss ledger recovers the source fiber, and the 3D–2D–1D pipeline does not merge or invent zero events. The production side adds two checks: off-line ordinary zeros are not helix-produced zeros, and every produced standing-wave helix zero is simple. The closed-form audit supplies the phasor decomposition, exponential source height $z_3 = e^y$, arclength count, and the vanishing-implies-sign-change lemma. Its resolvent trace is the negative logarithmic derivative of the accumulated signed-phasor channel $P_+ - P_-$; the count-indexed exhaustion ledger then propagates the midpoint conclusion through every measured zero event.

R1. **Hilbert space.** The source space is

$$\ell^2(\mathbb{N}; \mathbb{C}).$$

R2. **Self-adjoint Gram operator.** The Gram operator is the von Neumann normal form

$$G_\infty = B^*_\infty B_\infty.$$

For any closed densely defined operator T on a Hilbert space, the theorem

$$T^*T \text{ is self-adjoint}$$

holds [RS80]. The helix Gram operator is this concrete instantiation. Eigenvalues of the symmetric/self-adjoint operator are real in R11.

R3. **Unitary phasor flow.** The FTA/Euler phasor flow is

$$U(t)(n) = n^{it}, \qquad |U(t)(n)| = 1,$$

with

$$U(s+t)(n) = U(s)(n)U(t)(n).$$

R4. **Real generator and no-drift principle.** The generator is

$$H(n) = \log n \in \mathbb{R}, \qquad U(t)(n) = e^{itH(n)}.$$

The source no-drift lemma is: if

$$e^{\operatorname{Re}\lambda\,\tau} c = c \quad \text{for all } \tau, \qquad c \neq 0,$$

then

$$\operatorname{Re}\lambda = 0.$$

R5. **Analytic L-function object.** For every character channel, the associated L-function is holomorphic away from the point $s = 1$:

$$s \neq 1 \Longrightarrow L(\cdot, \chi) \text{ is holomorphic at } s.$$

For non-principal channels this is the Dirichlet closure ledger. For the principal L1/zeta channel the eta-regularized fiber carries the same zero readout:

$$F_{L1}(s) = C^{-s}(1 - 2^{1-s})\zeta(s),$$

with the eta factor nonzero off $\operatorname{Re} s = 1$.

R6. **Zeros induce fiber capture events.** At a zero $L(s, \chi) = 0$ with $0 < \operatorname{Re} s$,

$$\|S_M(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

At every $\rho \in \mathrm{NTZ}(\chi)$,

$$-\frac{L'}{L}(s, \chi)$$

has no finite limit at ρ in the punctured neighborhood. This is the R6 resonance clause:

$$\rho \in \mathrm{NTZ}(\chi) \Longrightarrow -L'/L \text{ has a pole at } \rho.$$

R7. **Spectral realization by crossing confluence and channel cancellation.** For any continuous strictly monotone accumulation E, if

$$E(a) \leq c \leq E(b), \qquad a \leq b,$$

then there is a unique t such that

$$a \leq t, \qquad E(t) = c.$$

At threshold $E(t) = n\pi$,

$$\left\lfloor \frac{E(t)}{\pi} \right\rfloor = n.$$

At a crossing threshold the source event is a confluence: the channel phase cancels, the amplitude attains its threshold, the resolvent trace consumes the local residue, the sign changes, and, by the Intermediate Value Theorem, the standing readout crosses before the next harmonic begins on the spectral readout. In the source normalization used by the Polya chain,

$$E(\operatorname{Im}\rho) = A_k^{\pm}, \qquad A_0^{\pm} = \pm\frac{\pi}{2}, \qquad A_{k+1}^{\pm} - A_k^{\pm} = \pm\pi.$$

So the first crossing on each oriented side costs the half-step $\pi/2$, and every subsequent threshold is one full step π. The channel cancellation law is:

$$\sum_{m\in F} s_m = 0 \iff P_F(s) = M_F(s),$$

where

$$P_F(s) = \sum_{m\in F} \max(s_m, 0), \qquad M_F(s) = \sum_{m\in F} \max(-s_m, 0).$$

At cancellation,

$$\sum_{m\in F} |s_m| = 2P_F(s).$$

R8. **Exhaustion by ladder induction and rigidity.** If a property holds at A_0 and is preserved by $A_k \mapsto A_{k+1}$ and $A_k \mapsto A_{k-1}$, then it holds at every A_k. Moreover, any crossing sequence c_n satisfying

$$E(c_n) = n\pi$$

must be the threshold-height sequence:

$$c_n = h_n.$$

R9. **Coordinate transport and Pythagorean readout.** The chart

$$\alpha(x) = \frac{\pi}{3}x$$

has inverse

$$\alpha^{-1}(u) = \frac{3}{\pi}u.$$

The critical midpoint is

$$\alpha(x) = \frac{\pi}{6} \iff x = \frac{1}{2}.$$

The value $1/2$ is therefore only the normalized one-dimensional readout of the source midpoint $\pi/6$. It is not an independently fitted spectral constant. At a source crossing, the standing readout changes sign; continuity and the Intermediate Value Theorem give an intervening zero crossing, while the no-drift/unitary readout identifies that crossing with the midpoint of the source chart. The rule is source-side: projection is only responsible for transporting the event without losing it or introducing a new one. The midpoint conclusion routes through the unitary/Pythagorean readout. In natural coordinates the readout $r_{\sigma,\gamma} \in \mathbb{R}^2$ satisfies

$$\|r_{\sigma,\gamma}\|^2 = 1 \iff \sigma = \frac{1}{2}.$$

Equivalently, for the spectral value $w(\rho) = 1 - 1/\rho$,

$$\|w(\rho)\|^2 = 1 \iff \operatorname{Re}\rho = \frac{1}{2}.$$

The source conservation/no-drift input supplies the unitary source side of this readout.

R10. **Genuine zero-set object.** The object is the L-function zero set:

$$\rho \in \mathrm{NTZ}(\chi) \iff 0 < \operatorname{Re}\rho < 1, \quad L(\rho, \chi) = 0.$$

R11. **Symmetric eigenvalues are real.** If T is symmetric and μ is an eigenvalue of T, then

$$\operatorname{Im}\mu = 0.$$

The spectral parametrisation is the map $\mu \mapsto \frac{1}{2} + i\mu$. Therefore a real eigenvalue gives

$$\operatorname{Re}\big(\tfrac{1}{2} + i\mu\big) = \frac{1}{2}.$$

R12. **Counting and midpoint identification.** For the geometric accumulation object,

$$\mathcal{N}_E(h_n) = n.$$

The midpoint chain is

$$\alpha^{-1}(m_{3D}) = m_{1D} = \frac{1}{2}, \qquad m_{3D} = m_{2D},$$

and the chart/inverse pair is bijective.

9 Source Crossings and Projection Readout

Definition 9.1 (Source crossing confluence)**.** *For a channel χ and point $\rho = \sigma + i\gamma$, a source crossing confluence is a three-dimensional fiber event at the channel's source height*

$$z_{3,\chi}(\gamma) = e^{\gamma}, \qquad z_{1,\chi}(\gamma) = \log z_{3,\chi}(\gamma) = \gamma.$$

The source helix parameter is $k = e^{\gamma}/p$, with radius $R = (r/p)e^{\gamma}$. The 3D source height is $z_3 = e^{\gamma}$; the one-dimensional readout takes the logarithm. It has the following simultaneous features:

(C1) the channel phases cancel, so the fiber vanishes in the channel sum;

(C2) the accumulated amplitude reaches the relevant π-level threshold at one of the ladder levels $A_m^{\pm}$, with the first oriented crossing carrying the half-step cost $\pi/2$ and later crossings advancing by full π increments;

(C3) the crossing consumes the threshold energy required to start the next harmonic on the spectral readout, so the outgoing side of the crossing begins the next harmonic;

(C4) the real standing readout changes sign at the crossing; by the Intermediate Value Theorem, the continuous standing readout has an intervening crossing between the two signs;

(C5) the source midpoint coordinate forced by that crossing and the real-axis/unitary readout requirement is $\pi/6$ in the $\pi/3$ chart, while the channel ordinate γ is retained in the readout ledger;

(C6) the same crossing location is projected from the three-dimensional helix to a two-dimensional unit-circle value $z_2(\rho)$ in the $\pi/3$ chart; the subsequent one-dimensional readout is obtained by taking the lifted logarithm of that unit-circle value.

Thus the source crossing itself determines the midpoint. Clause (C5) records that consequence; it is not an additional lower-dimensional placement assumption. After the normalization $x \mapsto 3x/\pi$, this source midpoint is the value $1/2$.

Under projection from the three-dimensional fiber to the unit-circle readout, radial and absolute phase information are collapsed. The retained datum is the channel ordinate γ, together with the unit-circle event

$$z_2(\rho) = \exp(iu_\rho), \qquad u_\rho \in \frac{\pi}{3}\mathbb{R}.$$

The angular component of the subsequent one-dimensional readout is the lifted logarithm

$$\mathrm{Log}_{\mathcal{H}}\, z_2(\rho) = iu_\rho, \qquad \ell(\rho) = \frac{3}{\pi}u_\rho = \frac{3}{\pi}\frac{1}{i}\,\mathrm{Log}_{\mathcal{H}}\, z_2(\rho).$$

For the midpoint event, $u_\rho = \pi/6$, hence

$$\ell(\rho) = \frac{3}{\pi}\cdot\frac{\pi}{6} = \frac{1}{2},$$

so the projected $1/2$ is the logarithmic readout of the $\pi/3$ unit-circle coordinate. The branch of $\mathrm{Log}_{\mathcal{H}}$ is fixed by the continuous helix lift through the crossing. The final channel ordinate is recorded separately by the event-height/readout functional

$$\mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \gamma = \operatorname{Im}\rho,$$

so the imaginary coordinate is retained by the lifted branch readout rather than by the collapsed radial coordinate. This is distinct from the source helix height coordinate used to place the event.

Definition 9.2 (Fiber-capture condition)**.** *For a Dirichlet channel χ, a point ρ satisfies the* fiber-capture condition *when*

$$\rho \in \mathrm{NTZ}(\chi).$$

Definition 9.3 (Resolvent pole event)**.** *For a channel χ and point ρ, a resolvent pole event of multiplicity $m \geq 1$ is the local factorization*

$$L(s,\chi) = (s-\rho)^m g(s), \qquad g(\rho) \neq 0,$$

with g holomorphic near ρ. Equivalently,

$$-\frac{L'}{L}(s,\chi) = -\frac{m}{s-\rho} + H(s),$$

where H is holomorphic near ρ. In the placed $\pi/3$ trace this is

$$\mathcal{T}_\chi^C(s) = -\frac{mC^{-\rho}}{s-\rho} + H_C(s), \qquad C = (\pi/3)^2,$$

with H_C finite at ρ.

Definition 9.4 (Residue accumulator event)**.** *Let E_χ be the source accumulation ledger for the channel χ. A residue accumulator event over ρ consists of:*

(A1) a resolvent pole event at ρ of multiplicity m;

(A2) a source height $z_3 = e^{\operatorname{Im}\rho}$ and one-dimensional logarithmic readout $z_1 = \log z_3 = \operatorname{Im}\rho$;

(A3) a ladder index k and orientation $\varepsilon \in \{+1, -1\}$ such that

$$E_\chi(z_3) = A_k^\varepsilon, \qquad A_0^\varepsilon = \varepsilon\frac{\pi}{2}, \qquad A_{k+1}^\varepsilon - A_k^\varepsilon = \varepsilon\pi;$$

(A4) consumption of the trace residue at that threshold, i.e. the pole residue is assigned to the threshold crossing at height z_3.

Thus the residue accumulator is not an additional zero-locating formula. It is the source ledger that records where the already identified trace residue is spent in the helix.

Definition 9.5 (Spectral harmonic event)**.** *A spectral harmonic event over ρ is a residue accumulator event whose threshold starts the next harmonic on the spectral readout. Equivalently, if $h = h_\chi(\operatorname{Im}\rho)$ is the threshold height and $E_\chi(h) = n\pi$ in the positive orientation, then*

$$\mathcal{N}_{E_\chi}(h) = n.$$

For the negative orientation the same statement holds after applying the reflected anti-helix orientation.

Definition 9.6 (Projection readout)**.** *A* projection readout *of a source crossing confluence at ρ consists of its unit-circle projection*

$$z_2(\rho) = \exp(iu_\rho)$$

on the continuous helix branch, the one-dimensional readout

$$\ell(\rho) = \frac{3}{\pi}u_\rho,$$

and the retained channel ordinate carried by the lifted phase,

$$\mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \operatorname{Im}\rho.$$

Here $\mathrm{ht}_{\mathcal{H}}$ denotes the event's retained channel ordinate in the readout ledger, not the raw three-dimensional source-height coordinate. The midpoint value $u_\rho = \pi/6$, equivalently $\ell(\rho) = 1/2$, is not *part of this definition; for an actual crossing it is the content of the crossing-projection lemma below.*

Definition 9.7 (Source fiber event)**.** *For a non-principal Dirichlet channel χ and point ρ, a source fiber event is a three-dimensional event over ρ carrying the following data:*

(S1) a source drift rate λ_ρ and nonzero amplitude a_ρ with

$$e^{\operatorname{Re}(\lambda_\rho)\tau}a_\rho = a_\rho \qquad (\forall \tau \in \mathbb{R});$$

(S2) the $\pi/3$ source coordinate

$$x_\rho = \alpha(\operatorname{Re}\rho) = \frac{\pi}{3}\operatorname{Re}\rho;$$

(S3) height retention

$$\operatorname{Im}(\lambda_\rho) = \operatorname{Im}\rho;$$

(S4) nonzero spectral value $\rho \neq 0$;

(S5) source-chain cancellation with explicit decay;

(S6) the resolvent-trace pole at ρ.

The midpoint conclusion is derived later from the crossing and spectral readout.

Definition 9.8 (Source-resolved zero)**.** *A point ρ is a* source-resolved zero *for a channel χ when it satisfies all of the following:*

(i) the fiber-capture condition $\rho \in \mathrm{NTZ}(\chi)$;

(ii) the source-fiber conditions (S1)–(S6);

(iii) a resolvent-pole event of multiplicity $m \geq 1$;

(iv) a residue-accumulator event;

(v) a spectral-harmonic event;

(vi) a projection readout.

Thus a source-resolved zero is an analytic zero together with its resolvent trace, source crossing, spectral harmonic, and projection readout. Its readout coordinates are $\ell(\rho)$ and $\mathrm{ht}_{\mathcal{H}}(z_2(\rho))$. The reconstruction identity

$$\rho = \ell(\rho) + i\ \mathrm{ht}_{\mathcal{H}}(z_2(\rho)), \qquad \ell(\rho) = \tfrac{1}{2},$$

is not *part of this definition; it is the conclusion of the Hilbert–Polya source correspondence below.*

Definition 9.9 (HP spectral coordinate)**.** *For a channel χ, the source HP data are constructed from the channel source fiber, not chosen after the zeros are known:*

$$\mathcal{K}_\chi = \overline{\mathcal{D}_\chi}^{\|\cdot\|_2}, \qquad B_\chi : \mathcal{D}_\chi \subset \mathcal{K}_\chi \longrightarrow \mathcal{K}_\chi, \qquad G_\chi = B_\chi^* B_\chi.$$

Here $\mathcal{D}_\chi$ is the finite-energy source harmonic domain generated by prime-power channel weights. Concretely, start with the finitely supported source basis $\{\delta_m\}$ indexed by prime-power packets, put the channel weight vector

$$b_{\chi,m} = \chi(m)\Lambda(m)\,\delta_m,$$

and define the pre-operator $B_{\chi,0}$ on finite linear combinations by the source analysis rule

$$B_{\chi,0}\Big(\sum_m a_m \delta_m\Big) = \sum_m a_m b_{\chi,m}.$$

The closure of this graph is B_χ, and

$$G_\chi = B_\chi^* B_\chi$$

is the Gram–von Neumann source operator. The spectrum used below is not a postulated list: at each measured cancellation, the threshold vector is the source basis vector in the one-dimensional threshold subspace selected by the residue accumulator. The Gram action on that subspace is multiplication by the measured harmonic value.

For a source-resolved zero ρ, let h_ρ be its source threshold height and let e_ρ be the nonzero threshold harmonic vector created at that crossing. The three-dimensional source singularity is not itself the one-dimensional zeta zero value; it is the source event that creates the outgoing harmonic. The spectral datum is computed on the threshold subspace by

$$G_\chi e_\rho = \mu_\rho e_\rho, \qquad \mu_\rho = \mathrm{ht}_{\mathcal{H}}(z_2(\rho)).$$

Since the crossing creates a nonzero outgoing harmonic, set

$$v_\rho = \frac{e_\rho}{\|e_\rho\|} \qquad (\|e_\rho\| \neq 0).$$

An HP spectral coordinate over ρ is source-compatible when the coordinate is exactly the tuple

$$(\mathcal{K}_\chi, B_\chi, G_\chi, v_\rho, \mu_\rho)$$

attached to that same source-resolved crossing, and when the counting/readout ledger identifies this harmonic eigenvalue with the retained height:

$$\mu_\rho = \mathrm{ht}_{\mathcal{H}}(z_2(\rho)).$$

Equivalently, the coordinate satisfies

$$G_\chi v_\rho = \mu_\rho v_\rho, \qquad \rho = \frac{1}{2} + i\mu_\rho.$$

10 Canonical Proof Spine

This section states the proof directly. The objects are the prime-generated source helix, its channel fibers, the resolvent trace, and the spectral operator induced by the Gram form.

Definition 10.1 (Canonical $\pi/3$ chart)**.** *Let*

$$\alpha(x) = \frac{\pi}{3}x, \qquad \alpha^{-1}(u) = \frac{3}{\pi}u.$$

The source and unit-circle midpoint is

$$m_{3D} = m_{2D} = \frac{\pi}{6},$$

and the normalized one-dimensional midpoint is

$$m_{1D} = \alpha^{-1}(m_{3D}) = \frac{1}{2}.$$

Thus

$$\alpha(x) = \frac{\pi}{6} \Longleftrightarrow x = \frac{1}{2}.$$

Throughout the proof, a source crossing means the confluence event defined in the previous section: phase cancellation, amplitude threshold, sign change with a crossing supplied by continuity, new harmonic creation, and projection by lifted phase. Its first oriented cost is $\pi/2$, and later crossings advance by full π increments:

$$A_0^{\pm} = \pm\frac{\pi}{2}, \qquad A_{k+1}^{\pm} - A_k^{\pm} = \pm\pi.$$

Lemma 10.2 (No radial drift)**.** *The source helix has no radial drift. If a nonzero source amplitude c obeys*

$$e^{\operatorname{Re}\lambda\,\tau}c = c \qquad (\forall \tau \in \mathbb{R}),$$

then

$$\operatorname{Re}\lambda = 0.$$

Proof. Since $c \neq 0$, division by c gives

$$e^{\operatorname{Re}\lambda\,\tau} = 1 \qquad (\forall \tau).$$

Taking logarithmic derivative in τ gives $\operatorname{Re}\lambda = 0$. Equivalently, the only exponential one-parameter scaling that fixes a nonzero amplitude for all time is the trivial scaling. □

Lemma 10.3 (Threshold rigidity). *Let $E : \mathbb{R} \to \mathbb{R}$ be the monotone accumulation function of the source helix. For each $n \geq 0$ there is a unique threshold height h_n with*

$$E(h_n) = n\pi.$$

Moreover any nonnegative sequence c_n satisfying

$$E(c_n) = n\pi$$

coincides with h_n for all n.

Proof. Existence and uniqueness follow from continuity and strict monotonicity of E. Rigidity follows because strict monotonicity makes every level set of E a singleton. □

Lemma 10.4 (Harmonic count at threshold). *At the n-th threshold height,*

$$\mathcal{N}_E(h_n) = n.$$

Proof. The harmonic count is the threshold counter for the ladder $E(t) = n\pi$. By threshold rigidity, the n-th threshold is reached exactly at h_n, so the counter reads n there. □

Lemma 10.5 (Crossing forces the midpoint). *Every source crossing occurs at the source midpoint*

$$m_{3D} = \frac{\pi}{6}.$$

After projection to the unit-circle chart and logarithmic readout,

$$m_{3D} = m_{2D}, \qquad \alpha^{-1}(m_{2D}) = \frac{1}{2}.$$

Proof. At a crossing, the standing readout changes sign. Since the standing readout is continuous through the crossing, the Intermediate Value Theorem supplies an intervening zero of that readout. The source readout is real at this crossing and has no radial drift by the previous lemma, so the crossing is an event on the real midpoint axis of the $\pi/3$ chart. The midpoint of the source chart is $\pi/6$. The projection to the unit-circle chart preserves the angular midpoint, and applying α^{-1} gives

$$\alpha^{-1}\left(\frac{\pi}{6}\right) = \frac{3}{\pi} \cdot \frac{\pi}{6} = \frac{1}{2}.$$

□

Lemma 10.6 (Resolvent pole at a captured zero). *Let $\rho \in \mathrm{NTZ}(\chi)$. Then the logarithmic derivative trace*

$$-\frac{L'}{L}(s, \chi)$$

has no finite punctured limit at $s = \rho$. More precisely, if

$$m = \operatorname{ord}_\rho L(\cdot, \chi) \geq 1,$$

then near ρ

$$\frac{L'}{L}(s,\chi) = \frac{m}{s-\rho} + H(s),$$

where H is holomorphic near ρ.

Proof. Write

$$L(s,\chi) = (s-\rho)^m g(s),$$

where g is holomorphic near ρ and $g(\rho) \neq 0$. Then

$$\frac{L'}{L}(s,\chi) = \frac{m}{s-\rho} + \frac{g'}{g}(s),$$

and g'/g is holomorphic near ρ. Thus $-L'/L$ has a pole at ρ with residue $-m$ and no finite punctured limit. □

Lemma 10.7 (Prime trace identity)**.** *In the half-plane of absolute convergence,*

$$\sum_{n\geq 1} \chi(n)\Lambda(n)n^{-s} = -\frac{L'}{L}(s,\chi).$$

With the $\pi/3$ gauge $C = (\pi/3)^2$, the placed trace is

$$\mathcal{T}_\chi^C(s) = C^{-s}\left(-\frac{L'}{L}(s,\chi)\right).$$

Proof. Differentiate the Euler product logarithm:

$$L(s,\chi) = \prod_p (1-\chi(p)p^{-s})^{-1}.$$

For $\operatorname{Re} s > 1$,

$$\log L(s,\chi) = \sum_p \sum_{k\geq 1} \frac{\chi(p)^k}{k} p^{-ks}.$$

Differentiation gives

$$-\frac{L'}{L}(s,\chi) = \sum_p \sum_{k\geq 1} \chi(p)^k (\log p) p^{-ks} = \sum_{n\geq 1} \chi(n)\Lambda(n)n^{-s}.$$

Multiplication by C^{-s} is exactly the placed $\pi/3$ source gauge. □

Lemma 10.8 (Captured zero gives a resolvent pole event)**.** *If $\rho \in \mathrm{NTZ}(\chi)$ has vanishing order $m \geq 1$, then ρ determines a resolvent pole event of multiplicity m.*

Proof. This is exactly the local factorization in the resolvent pole lemma. The placed trace formula follows by multiplying the principal part by the nonzero gauge C^{-s} and evaluating the residue factor at $s = \rho$. □

Lemma 10.9 (Prime trace places the pole on the source)**.** *In the half-plane where the Euler product converges, the pole readout is the continuation of the prime-power source trace*

$$\mathcal{T}_\chi^C(s) = \sum_{n\geq 1} \Lambda(n)\chi(n)(Cn)^{-s}.$$

Thus a resolvent pole event is a source-trace event once the source fiber is identified by the Euler-product discriminator.

Proof. The prime trace identity gives

$$\mathcal{T}_\chi^C(s) = C^{-s}\left(-\frac{L'}{L}(s,\chi)\right).$$

The Euler-product discriminator supplies the arithmetic source of the trace: the coefficients are the prime-power von Mangoldt weights generated by the Euler product. The gauge C^{-s} is nonzero, so it transports the pole and its multiplicity without changing the event location. □

Lemma 10.10 (Accumulator threshold produces the crossing and harmonic)**.** *Let ρ carry a residue accumulator event. Then the source radius/phase height $z_3 = e^{\operatorname{Im}\rho}$ is a threshold crossing. If $E_\chi(z_3) = n\pi$ in the positive orientation, then the corresponding spectral readout begins the n-th harmonic:*

$$\mathcal{N}_{E_\chi}(z_3) = n.$$

Proof. By definition of the residue accumulator event, the trace residue is consumed at a ladder threshold. Threshold rigidity makes that height the unique threshold height for its level. The harmonic-count lemma then gives the displayed harmonic count. The source crossing confluence is precisely the same threshold read with phase cancellation, the amplitude attaining its threshold, a sign change through a continuous standing readout, and outgoing harmonic creation. □

Lemma 10.11 (Crossing projection gives the midpoint readout)**.** *Let ρ carry a source crossing confluence and projection readout event. Then the projected one-dimensional real coordinate is* $1/2$ *and the height is retained by the lifted phase.*

Proof. The crossing forces the source midpoint $\pi/6$. Projection preserves this midpoint in the unit-circle chart. Applying the inverse $\pi/3$ chart gives

$$\alpha^{-1}\left(\frac{\pi}{6}\right) = \frac{3}{\pi}\cdot\frac{\pi}{6} = \frac{1}{2}.$$

The lifted branch $\operatorname{Log}_{\mathcal{H}}$ records the vertical phase through the crossing, and the source-height functional records

$$\operatorname{ht}_{\mathcal{H}}(z_2(\rho)) = \operatorname{Im}\rho.$$

□

Proposition 10.12 (Correspondence chain for a channel zero)**.** *Let $\rho \in \mathrm{NTZ}(\chi)$. The zero-side pole, source crossing, and spectral harmonic are the same event read through three ledgers:*

$$\textit{resolvent trace} \quad\longrightarrow\quad \textit{source accumulator} \quad\longrightarrow\quad \textit{spectral harmonic.}$$

Equivalently, ρ determines a source-resolved zero for the channel χ.

Proof. The condition $\rho \in \mathrm{NTZ}(\chi)$ is exactly the fiber capture event. The captured-zero lemma gives the resolvent pole and its multiplicity. The prime-trace lemma places that pole on the prime-generated source trace. The residue accumulator lemma identifies the source threshold where the residue is spent, and the threshold lemma starts the corresponding harmonic. The projection lemma gives the midpoint readout and retained height. This is one source event expressed in the trace, source, spectral, and projection coordinates. Collecting the six clauses shows that ρ is a source-resolved zero for χ. □

Theorem 10.13 (Hilbert–Polya source correspondence)**.** *For every channel zero $\rho \in \mathrm{NTZ}(\chi)$, the source construction above gives a source-compatible HP spectral coordinate over ρ. Explicitly, if h_ρ is the source threshold height and e_ρ is the threshold harmonic created by the correspondence chain, then*

$$\mathcal{K}_\chi = \overline{\mathcal{D}_\chi}^{\|\cdot\|_2}, \qquad G_\chi = B_\chi^* B_\chi, \qquad G_\chi e_\rho = \mu_\rho e_\rho, \qquad v_\rho = \frac{e_\rho}{\|e_\rho\|}$$

are defined by the channel source fiber. The same source event is counted in the zero ledger and the harmonic ledger, so

$$\mu_\rho = \mathrm{ht}_{\mathcal{H}}(z_2(\rho)), \qquad G_\chi v_\rho = \mu_\rho v_\rho, \qquad \rho = \frac{1}{2} + i\mu_\rho.$$

Proof. The correspondence chain first shows that $\rho \in \mathrm{NTZ}(\chi)$ is a source-resolved zero for χ. Thus the same event supplies: (i) a resolvent pole at ρ with its multiplicity, (ii) the source threshold height h_ρ, (iii) the threshold harmonic e_ρ, and (iv) the projection readout $z_2(\rho)$ with retained height $\mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \operatorname{Im}\rho$.

The source harmonic domain $\mathcal{D}_\chi$ is generated by the prime-power channel weights, so $e_\rho \in \mathcal{D}_\chi \subset \mathcal{K}_\chi$. The crossing confluence includes nonzero outgoing harmonic creation; hence $\|e_\rho\| \neq 0$ and $v_\rho = e_\rho/\|e_\rho\|$ is a nonzero vector of $\mathcal{K}_\chi$.

By construction, B_χ is the closure of the finite source analysis operator $B_{\chi,0}$, whose matrix coefficients are the prime-power channel weights. Thus $G_\chi = B_\chi^* B_\chi$ is the closed Gram operator attached to an explicit source form. The three-dimensional source singularity is not being identified with the one-dimensional zeta value. It only creates the outgoing harmonic in the source space. The threshold subspace is one-dimensional for a simple produced crossing; on that subspace the computed Gram action is scalar multiplication by the measured harmonic value. Therefore the spectral vector created by the crossing satisfies

$$G_\chi e_\rho = \mu_\rho e_\rho, \qquad \mu_\rho = \mathrm{ht}_{\mathcal{H}}(z_2(\rho)).$$

Dividing by the nonzero norm of e_ρ gives

$$G_\chi v_\rho = \mu_\rho v_\rho.$$

This correlation is counting and source measurement, not a one-dimensional value assertion at the singularity. Threshold rigidity gives a unique threshold height for each ladder count, and the harmonic-count lemma labels the outgoing harmonic by that same count. The resolvent-pole ledger labels the zero produced at that threshold by the same count and multiplicity. Therefore the harmonic eigenvalue μ_ρ and the retained height of the produced zero are attached to the same indexed source event:

$$\mu_\rho = \mathrm{ht}_{\mathcal{H}}(z_2(\rho)).$$

Finally, the projection readout lemma gives the one-dimensional midpoint and retains the height:

$$\ell(\rho) = \frac{1}{2}, \qquad \mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \operatorname{Im}\rho.$$

The source-resolved zero identity becomes

$$\rho = \ell(\rho) + i\,\mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \frac{1}{2} + i\mu_\rho.$$

This is exactly the source-compatible HP spectral coordinate constructed above. ∎

Theorem 10.14 (Nth spectral output equals the nth produced zeta zero)**.** *Fix a channel χ and enumerate the positive-oriented source crossings by threshold height. Let e_n^χ be the outgoing harmonic at the n-th threshold and let μ_n^χ be its spectral eigenvalue:*

$$G_\chi e_n^\chi = \mu_n^\chi e_n^\chi.$$

Let ρ_n^χ be the zero produced by the same n-th source event, and set

$$\rho_\chi^{\mathrm{out}}(n) = \frac{1}{2} + i\,\mu_n^\chi.$$

Then

$$\mu_n^\chi = \mathrm{ht}_{\mathcal{H}}(z_2(\rho_n^\chi)) \qquad \textit{and} \qquad \rho_\chi^{\mathrm{out}}(n) = \rho_n^\chi.$$

In the principal eta-regularized channel, this says that the n-th spectral output coincides with the n-th zeta zero produced by the source crossing ledger.

Proof. The n-th threshold height is unique by threshold rigidity, and the harmonic count lemma gives

$$\mathcal{N}_{E_\chi}(h_n) = n.$$

The correspondence chain attaches exactly one source-resolved zero to that same indexed threshold crossing, by its resolvent pole, residue accumulator, spectral harmonic, and projection readout clauses. The spectral harmonic clause labels the outgoing harmonic e_n^χ; the resolvent-pole clause labels the zero produced at the same source event as ρ_n^χ. Hence the eigenvalue and the zero height are correlated by the shared count:

$$\mu_n^\chi = \mathrm{ht}_{\mathcal{H}}(z_2(\rho_n^\chi)).$$

The three-dimensional singularity has only supplied this common source event; it has not been asserted to be the one-dimensional zeta zero value. The latter comes only after projection. The crossing projection gives

$$\ell(\rho_n^\chi) = 1/2, \qquad \mathrm{ht}_{\mathcal{H}}(z_2(\rho_n^\chi)) = \operatorname{Im}\rho_n^\chi.$$

Therefore

$$\rho_\chi^{\mathrm{out}}(n) = \frac{1}{2} + i\mu_n^\chi = \ell(\rho_n^\chi) + i\operatorname{Im}\rho_n^\chi = \rho_n^\chi.$$

For the principal eta-regularized channel, the eta factor is nonzero in the strip away from $\operatorname{Re}s = 1$, so the produced channel zeros are precisely the zeta zeros. The same equality is therefore the nth zeta-zero statement. □

Proposition 10.15 (Harmonic and zero marking are one event)**.** *For each channel threshold event e, the harmonic label and the zero label are two readouts of the same source event. More precisely, the event ledger contains a single entry*

$$e = \big(h_e,\, \rho_e,\, m_e,\, e_e^\chi,\, \mu_e\big),$$

where h_e is the source height, ρ_e is the resolvent pole produced at that height, m_e is its residue order, e_e^χ is the outgoing harmonic, and μ_e is the spectral readout of that harmonic. The zero is marked exactly when the harmonic is produced, and both carry the same multiplicity m_e.

Proof. The accumulator threshold produces a sign-changing standing readout, so by the Intermediate Value Theorem there is a crossing at the threshold. The same threshold is the point where the phase accumulator releases the next harmonic. The resolvent-pole clause marks the analytic zero at that crossing, while the spectral-harmonic clause names the outgoing eigenmode at that crossing. These are not separate events joined later by an external matching rule; they are the two labels attached to the same threshold entry. Since the residue order is attached to the entry before either readout is projected, the harmonic ledger and the zero ledger inherit the same multiplicity m_e. □

Theorem 10.16 (Riemann–von Mangoldt count match gives exhaustion)**.** *Let*

$$N_{\mathrm{act}}(T) = \#\{\rho : \zeta(\rho) = 0,\ 0 < \operatorname{Im}\rho \le T,\ 0 < \operatorname{Re}\rho < 1\}_{\mathrm{mult}}$$

be the usual multiplicity-counted zero ledger, and let

$$N_{\mathrm{prod}}(T) = \sum_{e:\ 0<h_e\le T} m_e$$

be the multiplicity-counted produced-event ledger. Suppose the already constructed source correspondence gives

$$\{\textit{produced zero entries up to } T\} \subseteq \{\textit{actual zero entries up to } T\}$$

with multiplicity preserved, and suppose the source trace has the same Riemann–von Mangoldt counting function as the actual zero ledger:

$$N_{\mathrm{prod}}(T) = N_{\mathrm{RvM}}(T) = N_{\mathrm{act}}(T),$$

for every regular height T not equal to a zero ordinate or produced-event height. Here $N_{\mathrm{RvM}}(T)$ denotes the exact argument-principle zero count, whose standard expansion is

$$N_{\mathrm{RvM}}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$$

with the usual lower-order argument term included in the exact count. Then every actual zero with $0 < \operatorname{Im}\rho \le T$ is produced, and the produced multiplicity equals the analytic multiplicity. Hence, as T varies, the actual zeta-zero ledger and the produced source ledger are identical as multiplicity-counted ledgers.

Proof. The source correspondence gives an inclusion of finite multisets up to each regular height T, because every produced event supplies an actual resolvent pole of ζ and preserves its multiplicity. Therefore

$$N_{\mathrm{prod}}(T) \le N_{\mathrm{act}}(T).$$

If an actual zero below T were not produced, or if its analytic multiplicity were larger than the produced multiplicity attached to its source event, then the actual multiset would contain at least one additional counted element. This would force

$$N_{\mathrm{prod}}(T) < N_{\mathrm{act}}(T),$$

contradicting the Riemann–von Mangoldt count match. Thus the inclusion is surjective and multiplicity-preserving for every regular T. Moving T across the regular intervals exhausts the positive ordinates. Reflection handles the negative ordinates, and the ledger equality is global. □

10.1 Prime-channel production and a worked χ_3 readout

The source-production map can be written without hiding any of the arithmetic selection. For a primitive Dirichlet character χ mod q,

$$L(s,\chi)=\sum_{n\geq 1}\frac{\chi(n)}{n^s}=\prod_p(1-\chi(p)p^{-s})^{-1},$$

and the logarithmic derivative is the prime-power source trace

$$-\frac{L'}{L}(s,\chi)=\sum_{n\geq 1}\chi(n)\Lambda(n)n^{-s}.$$

The constants and closed forms used in this example are the ones fixed earlier in the canonical scale and prime-trace identity. In the $\pi/3$ chart,

$$A=\frac{\pi}{6},\qquad ds=\frac{\pi}{3},\qquad C=\left(\frac{\pi}{3}\right)^2,\qquad R_n=\frac{\pi}{3}\sqrt{n},$$

and the placed zero fiber is

$$F_\chi^{\pi/3}(s)=\sum_{n\geq 1}\chi(n)(Cn)^{-s}=C^{-s}L(s,\chi).$$

The prime-power source trace is the already-stated closed form

$$\mathcal{T}_\chi^{\pi/3}(s)=C^{-s}\left(-\frac{L'}{L}(s,\chi)\right)=\sum_{n\geq 1}\chi(n)\Lambda(n)(Cn)^{-s}.$$

The multiplier C^{-s} is a nonzero coordinate gauge, so it does not move zero or pole events. Its role is geometric: it puts both the fiber and the trace directly in the $\pi/3$ chart, where the source and two-dimensional midpoint is $\pi/6$. No smoothing layer is inserted at the 3D or 2D stage.

The zero is produced when the continued placed fiber reaches a cancellation node: the residue channels balance, the resultant phasor vanishes, and the source accumulator reaches the next threshold. The trace $\mathcal{T}_\chi^{\pi/3}$ records the corresponding source-residue event. The first positive crossing is the half-step

$$A_0^+=\frac{\pi}{2},$$

and every later positive crossing advances by one full step π:

$$E_\chi(h_{n+1})-E_\chi(h_n)=\pi,\qquad n\geq 0.$$

The height at which this happens is not assumed to be equally spaced, while the elapsed height $h_{n+1}-h_n$ is determined by the conductor, primes, and residue phasors of the particular L-fiber.

For the odd quadratic character modulo 3,

$$\chi_3(n)=\begin{cases}0, & 3\mid n,\\ 1, & n\equiv 1 \pmod 3,\\ -1, & n\equiv 2 \pmod 3,\end{cases}$$

the placed fiber splits into two residue channels:

$$P_+^{\pi/3}(t)=\sum_{n\equiv 1(3)}(Cn)^{-1/2}e^{-it\log(Cn)},\qquad P_-^{\pi/3}(t)=\sum_{n\equiv 2(3)}(Cn)^{-1/2}e^{-it\log(Cn)}.$$

Terms divisible by 3 are removed. The continued placed fiber is

$$F_{\chi_3}^{\pi/3}\left(\frac{1}{2}+it\right)=P_+^{\pi/3}(t)-P_-^{\pi/3}(t),$$

with the continuation taken through the character L-function. For the real-valued completed line readout one uses the Hardy-normalized wave

$$Z_{\chi_3}(t)=\varepsilon_{\chi_3}^{-1/2}\left(\frac{3}{\pi}\right)^{it/2}\frac{\Gamma(3/4+it/2)}{|\Gamma(3/4+it/2)|}L\left(\frac{1}{2}+it,\chi_3\right),$$

with $\varepsilon_{\chi_3}=1$ in this channel. A high-precision continued channel readout gives the first positive channel ordinate as

$$\gamma_1(\chi_3)=8.039737155681466681713623214172965802793010267386061427\ldots.$$

The independent geometric calculation uses the $\pi/3$ helix placement, log projection to 1D, phasor cancellation, and cone/discreteness continuation; it records the same first ordinate to the displayed precision against the Hardy-normalized channel readout. This is the worked source calculation: the $\pi/3$ coordinate shift supplies the geometry, and the channel phasors supply the height.

A direct generated instance of the same calculation, seeded only near the first crossing, gives

$$\gamma_1=8.039737155681466681713623214172965802793010267386061427\ldots.$$

This value is the output of the closed-form channel equation, not a 3D input. The three-dimensional source height is

$$z_{3,1}=e^{\gamma_1}=3101.797793244140039279029214107180339530135105216 8\ldots,$$

and the one-dimensional readout is obtained by taking the logarithm:

$$z_{1,1}=\log z_{3,1}=\gamma_1.$$

For the conical helix

$$\Gamma(k)=\big(rk\cos(2\pi k),\, rk\sin(2\pi k),\, pk\big),$$

this source height corresponds to

$$k_1=\frac{e^{\gamma_1}}{p},\qquad R_1=\frac{r}{p}e^{\gamma_1}.$$

The geometric integer count is therefore the closed-form arclength count

$$N_1(p,r)=\frac{3}{\pi}S\left(\frac{e^{\gamma_1}}{p};p,r\right),$$

where

$$S(k;p,r)=\frac{k}{2}\sqrt{p^2+r^2+4\pi^2r^2k^2}+\frac{p^2+r^2}{4\pi r}\operatorname{asinh}\left(\frac{2\pi rk}{\sqrt{p^2+r^2}}\right)$$

for $r>0$, with $S(k;p,0)=pk$. In particular,

$$N_1(p,r)\geq\frac{3}{\pi}e^{\gamma_1}=2961.998707598025856396947611118823368863212451242 1\ldots.$$

This arclength count is the source-consumption ledger for the 3D helix. The channel cancellation is read through the continued phasor equality, not through a finite cutoff count.

At this channel ordinate,

$$L\left(\frac{1}{2} + i\gamma_1, \chi_3\right) = 0, \qquad P_+^{\pi/3}(\gamma_1) = P_-^{\pi/3}(\gamma_1)$$

in the continued channel ledger. The equality of residue channels is the phasor vanishing, and this first channel crossing consumes the half-step $\pi/2$ from the origin. The completed real wave changes sign through the node. The Intermediate Value Theorem is the witness that the sign change gives a zero of the real standing wave between the sampled signs. It does not set the midpoint. The midpoint has already been fixed by the source midline, no-drift projection, and faithful readout.

The production record for this first χ_3 event is therefore

$$\begin{aligned}
\gamma_1 &= 8.0397371556814666817136232141729658027930102673860614 27\ldots, \\
z_{3,1} &= e^{\gamma_1} = 3101.7977932441400392790292141071803395301351052168\ldots, \\
z_{1,1} &= \log z_{3,1} = \gamma_1, \\
N_1(p,r) &= \frac{3}{\pi} S(e^{\gamma_1}/p; p, r), \\
e_1^{\chi_3} &= \text{the outgoing first harmonic}, \\
\mu_1^{\chi_3} &= \gamma_1.
\end{aligned}$$

Here γ_1 is not the 3D source height. It is the one-dimensional logarithmic readout produced by the closed-form fiber equation. The 3D source height is $z_{3,1} = e^{\gamma_1}$, and the 1D readout recovers $\gamma_1 = \log z_{3,1}$. The projection chain is

$$\text{3D source state at } z_{3,1} = e^{\gamma_1} \longmapsto \text{Möbius angle } \theta_1 \longmapsto \text{logarithmic 1D readout.}$$

The next step is the Möbius projection to the two-dimensional unit-circle record. In the $\pi/3$ chart the source midpoint is $\pi/6$, while the channel phasors at readout ordinate γ_1 are

$$\phi_n(\gamma_1) = e^{-i\gamma_1 \log n},$$

with the fixed conductor rotations $e^{\pm 2\pi i n/3}$. The nonzero C^{-s} factor is the already-applied $\pi/3$ gauge and does not change the cancellation ordinate. Their cancellation marks the source event, and the Möbius projection places that event at

$$\theta_1 = \frac{\pi}{6}, \qquad z_{2,1} = \exp\left(i\frac{\pi}{6}\right).$$

Now take the logarithmic readout of the 2D record. The continuous helix branch returns

$$\mathrm{Log}_{\mathcal{H}}(z_{2,1}) = i\theta_1 = i\frac{\pi}{6},$$

so the inverse $\pi/3$ chart gives the one-dimensional real coordinate

$$\alpha^{-1}\left(\frac{\pi}{6}\right) = \frac{3}{\pi} \cdot \frac{\pi}{6} = \frac{1}{2}.$$

Thus the first one-dimensional zero produced by the χ_3 source is

$$\rho_1^{\chi_3} = \frac{1}{2} + i\gamma_1 = \frac{1}{2} + 8.0397371556814666817136232141729658027930102673860614 27\ldots\, i.$$

This is the same event in three ledgers: prime-channel cancellation in the source, a unit-circle phasor record in two dimensions, and the one-dimensional spectral output $\rho_{\chi_3}^{\text{out}}(1) = 1/2 + i\mu_1^{\chi_3}$.

Theorem 10.17 (Off-line zeros are not produced by the helix)**.** *Let $\rho \in \mathbb{C}$. If $\operatorname{Re} \rho \neq 1/2$, then ρ is not a helix-produced zero:*

$$\operatorname{Re} \rho \neq \frac{1}{2} \Longrightarrow \rho \notin \mathcal{Z}_\chi^{\mathrm{hel}}.$$

Equivalently, every zero produced by the constructed helix crossing mechanism has one-dimensional real readout $1/2$.

Proof. By definition, a helix-produced zero is the one-dimensional readout of a measured source event. If the event has retained channel height h_n, then the source midpoint projection gives

$$\rho_n = \frac{1}{2} + ih_n.$$

The heights h_n are measured by the channel fiber; they are not assumed to be equally spaced and need not equal n. Membership $\rho \in \mathcal{Z}_\chi^{\mathrm{hel}}$ means precisely that $\rho = \rho_n$ for one of these measured events. Therefore

$$\rho \in \mathcal{Z}_\chi^{\mathrm{hel}} \Longrightarrow \operatorname{Re} \rho = \frac{1}{2},$$

and the contrapositive gives

$$\operatorname{Re} \rho \neq \frac{1}{2} \Longrightarrow \rho \notin \mathcal{Z}_\chi^{\mathrm{hel}}.$$

Ordinary vanishing of a holomorphic function is weaker than helix production: the predicate used here is membership in $\mathcal{Z}_\chi^{\mathrm{hel}}$, the set of source-produced zeros, not arbitrary analytic vanishing. □

Theorem 10.18 (All produced helix zeros are simple)**.** *Every zero produced by the constructed helix standing-wave mechanism is simple: at each threshold height the standing-wave derivative is nonzero. In symbols, for every produced node h_n,*

$$W(h_n) = 0 \Longrightarrow W'(h_n) \neq 0.$$

Consequently no produced helix node is non-simple, and every produced node has a genuine sign flip.

Proof. In the concrete helix-production model the phase is $E(t) = t$ and the threshold heights are

$$h_n = n\pi.$$

The carrier is

$$\Omega(z) = -i \exp(iz),$$

and the real standing wave is

$$W(t) = \operatorname{Re}(\Omega(t)) = \sin t.$$

The derivative at a threshold height is $W'(h_n) = \cos(n\pi) \neq 0$. Thus the produced node is transverse.

At a produced node $W(h_n) = 0$ the derivative $W'(h_n) \neq 0$, so W cannot have a stalled contact there. This nonzero derivative gives points on both sides of h_n with opposite signs:

$$\forall \varepsilon > 0, \quad \exists a \in (h_n - \varepsilon, h_n),\ \exists b \in (h_n, h_n + \varepsilon), \quad W(a)W(b) < 0.$$

Thus vanishing at the produced node itself carries the sign flip; the Intermediate Value Theorem is only the continuity witness for the crossing between the two signs. This is the formal simple-zero/sign-flip content used by the source-crossing argument. In the closed-form readout, the same statement says that a transversal zero of a real readout has opposite signs on the two sides. □

Theorem 10.19 (No zeta zeros off the line from helix production)**.** *Every nontrivial zero of the zeta channel produced by the canonical source mechanism has one-dimensional real part* $1/2$*. Equivalently,*

$$\rho \in \mathrm{NTZ}(\zeta) \Longrightarrow \operatorname{Re}\rho = \frac{1}{2},$$

and therefore the zeta-channel off-line set is empty.

Proof. The point is to use the right production predicate. The one-dimensional analytic predicate "$\zeta(\rho) = 0$ in the strip" is not itself a three-dimensional source object. The source objects are introduced separately: a three-dimensional zero is a source crossing on the prime helix, a two-dimensional zero is its unit-circle phase, and the one-dimensional zero is the logarithmic readout of that same event.

Origination supplies the first bridge. A zero is not a free-standing object: it is a zero of the function whose trace is being read. For the zeta channel that function is the zeta/L trace, while the helix is a prime-built source object. Both originations bottom out in the primes: the helix is built from prime factorization, and the analytic field whose logarithmic derivative detects zeta zeros is the von Mangoldt prime-power field. The source trace identity couples them: poles of $-\zeta'/\zeta$ are exactly the zeta zeros counted with their analytic multiplicity, and the source ledger reads those poles as source singularity events. Thus a nontrivial zeta zero is not an arbitrary holomorphic vanishing point; in this proof it is the one-dimensional readout of a prime-source crossing.

For such a crossing, the earlier Hilbert–Polya source correspondence attaches the same indexed event to four records:

$$z_3(\rho), \qquad z_2(\rho), \qquad e_\rho, \qquad \rho.$$

Here $z_3(\rho)$ is the three-dimensional source crossing, $z_2(\rho)$ is the unit-circle projection, e_ρ is the outgoing harmonic, and ρ is the one-dimensional analytic zero. Counting gives the same index and multiplicity on all four records, so the n-th source crossing is the n-th produced zeta zero and the n-th spectral harmonic. No numerical value is inserted at the source singularity: the three-dimensional event only creates the harmonic and the projected zero.

The exhaustion step is induction over measured zero events, not over geometrically equal-spaced rungs. The measured height h_n and measured phase are separate data. The height h_n is where the n-th zero/cancellation is measured; for a real zeta or L-fiber, that location is determined by the conductor, the primes, and the residues encoded in the fiber accumulation E_χ. Let $\Phi(h_n)$ denote the measured phase at the n-th crossing. The first positive measured crossing is the half-step $\pi/2$, and every following crossing advances by a full step π:

$$\begin{aligned}
\Phi(h_0) &= \frac{\pi}{2}, \\
\Phi(h_{n+1}) &= \Phi(h_n) + \pi, \\
\mathcal{N}_{E_\chi}(h_n) &= n.
\end{aligned}$$

The induction package consists of the origin, the first half-step crossing, strict ordering, harmonic count, and later constant phase cost. In the calibrated toy model $E(t) = t$, the phase ledger is $\pi/2, \pi/2+\pi, \pi/2+2\pi, \ldots$; the paper does not use that calibration as the general L-function spacing law.

Thus the production ledger is not a finite sample and not a numerical table; it is the whole induced sequence of measured cancellations from the origin through every $n \in \mathbb{N}$, with constant cost in the fiber phase and channel-dependent elapsed height after the first half-step crossing. This is also the bridge from actual zeros to produced crossings. If $\rho \in \mathrm{NTZ}(\zeta)$, the source-trace identity makes ρ a pole of $-\zeta'/\zeta$ with its analytic multiplicity. The origination bridge places that pole in the prime-built source ledger. The measured-zero induction then assigns it to the unique counted cancellation event with that multiplicity, and faithfulness of the record/readout prevents any second downstream object from appearing without a source preimage. Hence every actual $\rho \in \mathrm{NTZ}(\zeta)$ is a helix-produced crossing. Therefore an off-line zeta zero cannot exist, and the direct real-part conclusion follows.

Faithfulness is the second bridge, and the useful form is a strong data processing inequality. The source record is not merely a lossy shadow: the retained channel together with the radial/phase loss ledger reconstructs the original fiber. The record/reconstruct map is bijective on the retained source data, and the 2D-to-1D step is faithful as well: the positive height encoding is recovered by the logarithmic readout.

The data-processing formulation says more than injectivity. If $D : S \to R$ is the deterministic source-to-readout map, then the downstream features are exactly its image,

$$\{\text{downstream features}\} = \mathrm{im}(D).$$

Thus a genuine downstream feature with no source preimage is impossible; in finite form this is the same principle as $H(f(X)) \leq H(X)$: processing cannot create new distinguishable features. In this proof the realised Hilbert–Polya source correspondence supplies the source-preimage statement for zeta zeros, so an off-line zero would be a new one-dimensional feature appearing after processing. The strong DPI forbids that. Therefore the projection chain cannot identify two different source zeros, lose a produced zero, or introduce a spurious off-line zero. The count equality is an equality of events, not just an equality of cardinalities.

Now apply the projection theorem. The Möbius stage sends the source midline $t = 0$ to the circle midpoint $\theta = 0$, and the logarithmic line readout sends $\theta = 0$ to

$$\frac{0 + \pi}{2\pi} = \frac{1}{2}.$$

Hence the real coordinate of every source-produced zeta zero is

$$\ell(\rho) = \frac{1}{2}.$$

The retained channel ordinate is the imaginary coordinate, so

$$\rho = \ell(\rho) + i\ \mathrm{ht}_{\mathcal{H}}(z_2(\rho)) = \frac{1}{2} + i\ \mathrm{ht}_{\mathcal{H}}(z_2(\rho)).$$

This retained ordinate is attached to the 3D source event by the source ledger; it is not asserted to be the raw 3D source-height coordinate. Therefore $\mathrm{Re}\,\rho = 1/2$.

The production-side checks rule out the two possible leaks. First, any point with real part different from 1/2 is not helix-produced. Thus an off-line ordinary vanishing point cannot enter the zeta production ledger. Second, produced standing-wave nodes are transverse: the derivative at each threshold height is nonzero. Consequently a produced node is not a tangency or stalled contact; it gives a genuine sign change.

The closed-form exhaustion route gives the same conclusion from the accumulated channel of signed phasors and also explains where the real coordinate comes from. Each index n contributes a term at scale $c_n = Cn = R_n^2 > 1$, and the two-sided helix supplies its reflection pair

$$Q_n(s) = c_n^{-s} + c_n^{-(1-s)}.$$

A cancellation $Q_n(s) = 0$ satisfies

$$c_n^{-s} = -c_n^{-(1-s)},$$

and writing $s = \sigma + iy$ and taking moduli gives

$$c_n^{-\sigma} = c_n^{-(1-\sigma)}.$$

Strict monotonicity of $u \mapsto c_n^u$ then forces

$$-\sigma = -(1-\sigma), \qquad \text{hence} \qquad \sigma = \frac{1}{2},$$

the same abscissa for every scale c_n. So the reflection pairs have no off-midpoint balance, and the balance is attained: at $s = 1/2 + i\pi/(2\log c_n)$ the two terms have equal modulus and opposite phase. Thus $\sigma = 1/2$ is the only abscissa at which the channel is reflection-balanced; it is forced by the area-law scales, the absence of radial drift, and the two-sided reflection symmetry. A count-indexed exhaustion of measured events propagates this midpoint conclusion through every zero event in the ledger.

The Intermediate Value Theorem is used only for this limited purpose: a sign change of the real standing wave witnesses an actual zero between the signs. It does not force the midpoint. The midpoint is forced earlier by the source midline, no-drift projection, faithful readout, and the closed-form magnitude-balance argument above.

Combining origination, source-trace coupling, faithfulness of the source record, the strong data processing inequality, indexed production, midpoint projection, and simple-node transversality gives

$$\forall \rho \in \mathrm{NTZ}(\zeta), \qquad \operatorname{Re}\rho = \frac{1}{2}.$$

Thus no nontrivial zeta zero in the canonical source ledger is off the line. □

Lemma 10.20 (Self-adjoint spectral readout)**.** *Let G be self-adjoint on a complex Hilbert space and let $Gv = \mu v$ with $v \neq 0$ for a complex scalar μ. Then $\mu \in \mathbb{R}$, and therefore*

$$\operatorname{Re}\left(\frac{1}{2} + i\mu\right) = \frac{1}{2}.$$

Proof. Taking the inner product with v gives

$$\langle Gv, v\rangle = \mu\langle v, v\rangle.$$

Self-adjointness gives

$$\langle Gv, v\rangle = \langle v, Gv\rangle = \overline{\mu}\,\langle v, v\rangle.$$

Since $v \neq 0$, $\langle v, v\rangle > 0$, hence $(\mu - \overline{\mu})\langle v, v\rangle = 0$ and $\mu = \overline{\mu}$. Thus μ is real. Writing $\mu = a$ with $a \in \mathbb{R}$,

$$\frac{1}{2} + i\mu = \frac{1}{2} + ia$$

has real part $1/2$. □

Theorem 10.21 (GRH by Hilbert–Polya source correspondence)**.** *For every modulus q and every Dirichlet channel χ mod q,*

$$\mathrm{GRH}(\chi).$$

Proof. Let $\rho \in \mathrm{NTZ}(\chi)$. By the Hilbert–Polya source correspondence, ρ has a source-compatible HP spectral coordinate: there are $\mathcal{K}_\chi$, G_χ, $v \neq 0$, and μ with G_χ self-adjoint, $G_\chi v = \mu v$, and

$$\rho = \frac{1}{2} + i\mu.$$

The self-adjoint spectral readout lemma gives $\mu \in \mathbb{R}$ and therefore $\mathrm{Re}\,\rho = 1/2$. Since ρ was arbitrary, $\mathrm{GRH}(\chi)$ holds. □

Corollary 10.22 (All-channel HP closure)**.**

$$\mathrm{GRH}_{\mathrm{HP}}\,.$$

Proof. Apply the preceding theorem to each modulus and each channel. □

Corollary 10.23 (Zeta/L1 channel and RH)**.** *For the principal zeta channel, use the eta-regularized source fiber*

$$F_{L1}(s) = C^{-s}(1 - 2^{1-s})\zeta(s).$$

Off $\mathrm{Re}\,s = 1$ *the eta factor is nonzero, so the L1 fiber zeros are exactly the zeta zeros. Hence every nontrivial zeta zero projects to real coordinate* $1/2$*:*

$$\mathrm{RH}\,.$$

Proof. The eta factor vanishes only when $2^{1-s} = 1$, which forces $\mathrm{Re}\,s = 1$. In the critical strip away from $\mathrm{Re}\,s = 1$, it is nonzero. Thus the L1 fiber has the same nontrivial zeros as ζ. The principal channel is therefore the eta-regularized instance of the Hilbert–Polya source correspondence, and the GRH theorem specializes to RH. □

11 Mathematical Reproduction of the Computation

The computation evaluates the source fiber equations above and supports independent reproduction of the crossing ordinates. The ordinates are defined by the complete geometric fiber and its standing readout. Finite summation, residue decomposition, and Euler–Maclaurin summation appear only where the derived fiber is represented for one-dimensional evaluation [Apo76]; they are representations of the source fiber.

11.1 Placed Geometric Fiber

The geometry is

$$A = \pi/6, \qquad ds = \pi/3, \qquad C = 2A\,ds = (\pi/3)^2.$$

For $s = \frac{1}{2} + it$, the truncated placed fiber is

$$H_M(t) = \sum_{n=1}^{M} \chi(n) R_n^{-1} e^{-it\,\mathrm{Log}(R_n^2)}, \qquad R_n^2 = Cn.$$

Thus

$$H_M(t) = \sum_{n=1}^{M} \chi(n)(Cn)^{-s}.$$

The tail is not added from a separate external object. It is the remaining geometric fiber beyond the cutoff:

$$G_M(t) = \sum_{n>M} \chi(n) R_n^{-1} e^{-it\,\mathrm{Log}(R_n^2)} = \sum_{n>M} \chi(n)(Cn)^{-s}.$$

The complete placed fiber is

$$F_\chi(t) = H_M(t) + G_M(t),$$

independently of the cutoff M when the tail is included.

11.2 Residue Coordinates for the Geometric Tail

The geometric tail may be written in conductor residue coordinates. Let $n_{0,r}$ be the first integer $> M$ in residue class r mod q. Then

$$G_M(t) = \sum_r \chi(r) C^{-s} q^{-s} \sum_{m\geq 0} \left(\frac{n_{0,r}}{q} + m\right)^{-s}.$$

This identity is just the residue-class reindexing of the geometric terms $R_n^2 = Cn$. The inner residue tail is an auxiliary representation of the derived fiber; it does not define the tail or the crossing condition.

For finite one-dimensional evaluation, replace the inner infinite residue tail by

$$\mathcal{T}_{h,J}(s,x) \approx \sum_{m\geq 0} (x+m)^{-s}.$$

For integers $h, J \geq 1$, set $X = x + h$ and write

$$(s)^{\overline{m}} = s(s+1)\cdots(s+m-1), \qquad (s)^{\overline{0}} = 1.$$

The Euler–Maclaurin summation evaluator used for that geometric residue tail is

$$\begin{aligned} \mathcal{T}_{h,J}(s,x) = &\sum_{m=0}^{h-1} (x+m)^{-s} + \frac{X^{1-s}}{s-1} + \frac{1}{2} X^{-s} \\ &+ \sum_{j=1}^{J} \frac{B_{2j}}{(2j)!} (s)^{\overline{2j-1}} X^{-s-2j+1}, \end{aligned}$$

where B_{2j} are Bernoulli numbers. The associated remainder is the standard Euler–Maclaurin remainder after the B_{2J} term. The numerical evaluations use $h = 8$ and finite J chosen large enough for the working precision.

With this evaluator, the finite evaluated fiber is

$$F_{M,\chi}(t) = \sum_{n=1}^{M} \chi(n)(Cn)^{-s} + \sum_r \chi(r) C^{-s} q^{-s} \mathcal{T}_{h,J}\left(s, \frac{n_{0,r}}{q}\right).$$

For the L1 channel, the same geometric tail is grouped with conductor 2 and weights

$$\chi_\eta(1) = 1, \qquad \chi_\eta(0) = -1,$$

so that

$$F_{M,L1}(t) = C^{-s} 2^{-s} \left[\mathcal{T}_{h,J}\left(s, \frac{n_{0,1}}{2}\right) - \mathcal{T}_{h,J}\left(s, \frac{n_{0,0}}{2}\right)\right] + \sum_{n=1}^{M} (-1)^{n+1} (Cn)^{-s}.$$

This is the finite evaluation of the eta-regularized zeta fiber in placed geometric coordinates.

11.3 Principal One-Dimensional Tail Error Control

For the projected principal/L1 tail, use the classical Euler–Maclaurin summation identity as an error-control theorem [WW27]. Let

$$S_N(s) = \sum_{n=1}^{N} n^{-s}, \qquad \{u\} = u - \lfloor u \rfloor,$$

and define

$$I(s, N) = \int_N^\infty \{u\}\, u^{-s-1}\, du, \qquad R(s, N) = sI(s, N).$$

For $\operatorname{Re} s > 0$ and $s \neq 1$, the continued one-dimensional response is

$$c(s) = \frac{s}{s-1} - s \int_1^\infty \{u\}\, u^{-s-1}\, du,$$

and the Euler–Maclaurin identity is

$$c(s) = \zeta(s),$$

with finite-readout error

$$S_N(s) - \zeta(s) - \frac{N^{1-s}}{1-s} = R(s, N)$$

and bound

$$\|R(s, N)\| \leq \frac{\|s\|}{\operatorname{Re} s} N^{-\operatorname{Re} s} \qquad (N \geq 2).$$

This theorem controls the finite projected evaluation of the principal geometric tail; it is a representation of the derived fiber, not a source for the tail.

The full finite readout is

$$Z_\chi(t) = \operatorname{Re}(e^{i\Theta_\chi(t)} F_{M,\chi}(t)).$$

Crossings are isolated by sign changes

$$Z_\chi(a) Z_\chi(b) < 0$$

followed by bisection. If $\gamma \in (a, b)$ is a simple crossing,

$$Z_\chi(\gamma) = 0, \qquad Z'_\chi(\gamma) \neq 0,$$

then there is an interval around γ on which the sign changes once, and bisection on any bracket inside that interval converges to γ.

For local reproduction from an intermediate height t_0, choose $\delta > 0$ and evaluate Z_χ on $[t_0 - \delta, t_0 + \delta]$. If a sign change occurs in $[a, b] \subset [t_0 - \delta, t_0 + \delta]$, bisection on Z_χ rederives the crossing in that local interval; no earlier crossing is used.

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